

Teaching Guide

All 17 modules in one printable volume

For every topic across the course, this volume includes the science to teach, the recommended approach to learning that material, and why it matters to your students' future patients. Each module is the companion to its student-facing video lesson and pre-work hub. Print front to back or pull single modules as needed.

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Dr. Sharilyn Rennie

Module 1: Foundations of Anatomy and Physiology

Topics: Levels of Organization, Anatomical Terminology and Body Regions, Homeostasis and Feedback Loops

TOPIC 1 OF 3 . WEEK 1 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Levels of Organization

From atoms to organism: how complexity scales and where new function emerges.

The Science

The hierarchy is a scaffold, not a list. The point is that every level builds on the one below and that function emerges as complexity rises. A single sodium atom does nothing interesting. Pack billions of them into the right gradient across a membrane and you get an action potential. Pack action potentials into the right wiring and you get a thought. The lesson is that biology lives in the relationships between parts, not in the parts themselves.

Anchor the eleven organ systems by what they do. Integumentary protects. Skeletal supports. Muscular moves. Nervous and endocrine communicate (one fast, one slow). Cardiovascular and lymphatic transport. Respiratory and digestive bring in oxygen and nutrients; urinary and respiratory remove waste. Reproductive perpetuates. Students who can recite eleven systems but cannot say what each is FOR will not retain the rest of the course.

Drop in two specific examples to make emergence concrete: cardiac contraction depends on intercalated discs (cell-level structure) creating a functional syncytium (tissue-level behavior) producing coordinated ejection (organ-level function). And gas exchange depends on the alveolar wall being one cell thick (cellular) so that diffusion distance is minimal (tissue), so that gas exchange is fast enough (organ) to support whole-body metabolism.

Teaching This Topic

Before the video (drawing prompt)

Ask students to draw the six levels as nested boxes and write one example per level from a body part of their choosing (heart, skin, etc.). They have to commit before the video gives them the answer.

Common misconceptions to address

- Students confuse 'organ system' with 'organ.' The kidney is an organ; the urinary system is the kidney plus ureters, bladder, urethra.
- Students think the levels are a one-time list to memorize. They are a recurring framework you will reach for throughout the entire course.
- Cells are often pictured as identical building blocks. Emphasize that there are hundreds of cell types, each shaped for its job.

Order of operations (what to teach first)

Levels first (hierarchy), then organ systems (with one-sentence functions), then emergence (with two concrete examples). Do not start with the organ system list. The hierarchy is the load-bearing concept.

Self-test prompts

- Without notes, list the six levels of organization in order.
- Pick three organ systems and write one sentence describing each one's job.
- Explain what 'emergent property' means using your own example.

Why This Matters to Your Patients

When your patient has a problem, you must locate it at the right level. A drug works at the chemical or cellular level but the patient experiences it at the organism level. Treatment-resistant hypertension might be a chemical problem (a receptor mutation), a cellular problem (renal cells malfunctioning), an organ problem (kidney damage), or a system problem (RAAS dysregulation). Each level demands a different intervention. Nurses who can move fluently between levels catch things others miss: the patient whose 'fatigue' turns out to be cellular (mitochondrial dysfunction from a statin) rather than psychological.

TOPIC 2 OF 3 . WEEK 1 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Anatomical Terminology and Body Regions

Anatomical position, directional terms, body planes, cavities, and the four abdominal quadrants.

The Science

Anatomical position is the reference frame. Every directional term, every plane, every regional name assumes the patient is standing in this position. If your patient is lying face down on the OR table, the surgeon does not say 'the kidney is above the bladder' from the patient's current orientation. They say 'superior to the bladder' from anatomical position. This consistency is the entire point.

Teach the directional terms as pairs, because they are always relative: superior versus inferior, medial versus lateral, proximal versus distal. Distal and proximal only make sense relative to an origin (usually the trunk for limbs). The wrist is distal to the elbow but proximal to the fingers. Students who chant directional terms in isolation miss this.

The three planes and the body cavities give you the vocabulary to describe any imaging study. A transverse CT slice through the abdomen will show kidneys, vertebra, and large vessels in cross-section. A sagittal MRI of the brain will show the corpus callosum lengthwise. The nine abdominal regions are more precise than the four quadrants and worth the small extra investment.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a stick figure in anatomical position and label as many directional pairs as they can. Then have them draw a transverse, sagittal, and frontal section through the figure.

Common misconceptions to address

- Students forget that anatomical position is a fixed reference and try to update terms when the patient moves.
- Confusion between dorsal/ventral and posterior/anterior. They are synonyms in humans; in four-legged animals they differ.
- Students think 'midsagittal' and 'sagittal' are the same. Midsagittal is the unique plane through the middle; any parallel plane is sagittal.

Order of operations (what to teach first)

Position first, then directional terms (in pairs), then planes (with imaging examples), then cavities and regions. The vocabulary loads incrementally.

Self-test prompts

- Without notes, give five pairs of directional terms.
- Sketch a transverse, frontal, and sagittal plane through the body.
- Place your hand on the upper right region of your abdomen and name the region and likely organs.

Why This Matters to Your Patients

This is the vocabulary every clinical report uses. A radiology read will say 'mass in the right upper quadrant, anterior to the kidney.' You cannot follow the report without the language. A nursing assessment will document pain 'lateral to the umbilicus, radiating distally down the right leg.' This is not academic vocabulary; it is the minimum standard for clinical communication. Sloppy terminology causes real handoff errors. A patient with right lower quadrant pain has a very different workup than one with left lower quadrant pain.

TOPIC 3 OF 3 . WEEK 1 . 8 CARDS (4 DOK 1 / 2 DOK 2 / 2 DOK 3)

Homeostasis and Feedback Loops

Set points, the four-part negative feedback loop, and why positive feedback is the rare exception.

The Science

Homeostasis is the deepest concept in physiology. Almost every system you will study exists to regulate something. Make this load-bearing. The four parts of a negative feedback loop, stimulus, receptor, control center, effector, are a template you can lay over thermoregulation, blood glucose, blood pressure, blood pH, osmolarity, and calcium balance. Once they have the template, the rest of the course is variations on a theme.

Negative feedback opposes change. Positive feedback amplifies it. Positive feedback is rare in physiology because it is dangerous. The body only uses it when it needs to drive to an end point quickly (childbirth, clotting) and then shut it off. Anytime students see a system go to extremes, suspect a broken negative feedback loop or an inappropriately running positive one.

The set point is the target. Diseases often reset the set point rather than destroy the feedback machinery. Fever is the classic example: the hypothalamus has been told by pyrogens to defend 39 degrees instead of 37, and the entire feedback system dutifully drives toward the new target. The patient shivers (effector response) and the skin vasoconstricts (effector response) until they hit the new set point. Then they feel hot, because their body is hot. Get this and you understand a huge fraction of clinical physiology.

Teaching This Topic

Before the video (drawing prompt)

Ask students to draw a negative feedback loop for body temperature: arrows, labels, where the receptor is, where the effector is. Pre-commitment makes the feedback diagram stick.

Common misconceptions to address

- Students think 'negative feedback' is bad. Reframe: it is stabilizing.
- Students think positive feedback is rare because it is unimportant. It is rare because it is risky, but where it occurs it is essential.
- The set point is confused with the current value. They are different. The set point is the target; the current value is what the receptor is measuring.

Order of operations (what to teach first)

Definition, then loop components (with one shared example, like temperature), then negative versus positive contrast, then set point disturbances (fever). The loop components are the conceptual core; everything else hangs on them.

Self-test prompts

- From memory, list the four components of a negative feedback loop.
- Identify the receptor, control center, and effector in the baroreflex.
- Explain in your own words why fever feels cold at the start and hot at the end.

Why This Matters to Your Patients

Almost every disease you will see is a homeostatic failure. Diabetes is broken glucose homeostasis. Hypertension is broken blood pressure homeostasis. Acidosis and alkalosis are broken pH homeostasis. Heart failure breaks blood pressure, then volume, then respiratory homeostasis in a cascade. Nurses live in this world: every set of vitals is a homeostatic readout. When something is out of range you are not just noting numbers; you are asking which loop is failing and which compensation is firing. A tachycardic, hypotensive patient is showing you their baroreflex trying to save their blood pressure. Reading vitals as feedback loops, not just values, changes the level of care you give.

Module 2: The Cell

Topics: Cell Structure and Organelles, Membrane Transport

TOPIC 1 OF 2 . WEEK 1 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Cell Structure and Organelles

Plasma membrane, nucleus, and the major cytoplasmic organelles that do the work of the cell.

The Science

The plasma membrane is the boundary between alive and not-alive at the smallest scale. Phospholipid bilayer with embedded proteins. Selectively permeable: the cell decides what comes in and what goes out. Surface proteins are receptors, channels, transporters, and identity markers. If you teach nothing else about cells, teach the membrane. Everything in physiology happens across membranes.

Organelles divide labor. The nucleus stores DNA. The nucleolus inside makes ribosomal RNA. Free ribosomes make cytoplasmic proteins; rough-ER ribosomes make secreted and membrane proteins. The Golgi modifies and sorts proteins from the ER. Mitochondria are the ATP factories, with their own DNA (legacy of bacterial ancestry) and double membrane. Lysosomes digest debris. Peroxisomes detoxify reactive oxygen species and metabolize fatty acids.

Specialization is the story. A pancreatic acinar cell that secretes massive amounts of digestive enzymes has enormous rough ER. A cardiac myocyte that contracts non-stop is packed with mitochondria. A red blood cell, which just needs to carry hemoglobin, ditches its nucleus and most organelles entirely. Organelle abundance maps onto function; this is a question students will see again and again.

Teaching This Topic

Before the video (drawing prompt)

Ask students to draw a cell with the membrane and label what each major organelle does. They commit, then the video corrects them.

Common misconceptions to address

- Cells are pictured as identical generic blobs. Reality: hundreds of cell types, each shaped for function.
- Mitochondria are 'powerhouses' in cliche. Better: ATP factories, with their own DNA, and they sense and respond to the cell's energy state.
- Students confuse the ER with the Golgi. ER makes; Golgi modifies and ships.

Order of operations (what to teach first)

Membrane first (load-bearing), then nucleus, then the protein-secretion pipeline (ribosome to rough ER to Golgi), then mitochondria, then the cleanup crew (lysosomes, peroxisomes).

Self-test prompts

- List the organelles a pancreatic acinar cell would have in abundance and why.
- Why are red blood cells unusual in lacking a nucleus?
- What is the difference between rough ER and smooth ER in function?

Why This Matters to Your Patients

Mitochondrial diseases cause exercise intolerance, muscle weakness, and neurodegeneration because high-demand tissues fail first when ATP supply drops. Lysosomal storage diseases (Tay-Sachs, I-cell) accumulate undigested debris in cells, producing progressive organ failure. Drugs that target the plasma membrane, ion channels, receptors, transporters, are the backbone of pharmacology. A patient on a statin has had hepatic mitochondria tweaked. A patient on chemotherapy is having their cell division and protein synthesis machinery hit hard. Cellular biology is the level at which most modern medicine acts.

TOPIC 2 OF 2 . WEEK 1 . 8 CARDS (4 DOK 1 / 2 DOK 2 / 2 DOK 3)

Membrane Transport

Passive vs active, simple vs facilitated, primary vs secondary, plus bulk transport.

The Science

Anything that moves into or out of a cell goes either down its gradient (passive, free) or against its gradient (active, costs ATP). That single distinction organizes the entire topic.

Passive: simple diffusion through the bilayer (small, nonpolar molecules; O₂, CO₂, steroid hormones). Facilitated diffusion through channels or carriers (polar molecules and ions). Osmosis is water moving down its concentration gradient through aquaporins. All of these are 'free' to the cell. They use the energy already in the gradient.

Active: primary active transport burns ATP directly (the Na/K ATPase is the prototype, pumping 3 Na out and 2 K in per ATP). Secondary active transport uses the gradient set up by primary pumps (the Na-glucose symporter in the small intestine drags glucose in against its gradient by piggybacking on the Na coming in down its gradient). Build the conceptual chain: Na/K ATPase establishes the Na gradient; everything else in the cell uses that gradient as currency. This is why ouabain (a Na/K pump inhibitor) ultimately disables glucose absorption from the gut, even though it never touches the glucose transporter.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a cell membrane and predict which of these molecules cross passively versus require help: O₂, glucose, Na, water, cholesterol, K. Then check.

Common misconceptions to address

- Students think active transport means 'fast.' It means 'against the gradient.'
- Students lump all transport into 'osmosis.' Osmosis is specifically water.
- Aquaporins are sometimes thought of as exotic. They are everywhere; without them, water movement would be much too slow for physiology.

Order of operations (what to teach first)

Frame the question (down or up the gradient?), then passive types, then active types, then the Na/K ATPase as the master gradient-builder.

Self-test prompts

- Without notes, list two molecules that cross by simple diffusion and two that need help.
- Explain why blocking the Na/K pump eventually stops glucose absorption.
- Predict what happens to a red blood cell in pure water.

Why This Matters to Your Patients

Diuretics work by manipulating tubular transporters. Furosemide blocks the NKCC2 cotransporter in the loop of Henle, causing massive Na and water loss. Spironolactone blocks aldosterone (which controls Na reabsorption). IV fluids are designed around osmotic principles: isotonic saline keeps cells stable, hypotonic fluid drives water into cells (treating dehydration), hypertonic fluid pulls water out (treating cerebral edema). Cholera kills by hijacking the intestinal Cl channel, driving massive secretion. Cystic fibrosis is a defective Cl channel. Once you see the body as a network of transport mechanisms, much of clinical pharmacology becomes legible.

Module 3: Tissues

Topics: Epithelial Tissue Classification, Connective Tissues, Muscle and Nervous Tissue Overview

TOPIC 1 OF 3 . WEEK 2 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Epithelial Tissue Classification

Naming epithelia by layer and cell shape; where each type lives and what it does.

The Science

Epithelia cover and line. They are everywhere there is a body surface, external (skin) or internal (gut lumen, blood vessel, alveolus). Their job is barrier, exchange, secretion, or absorption, depending on location. They are avascular; nutrients come from the underlying connective tissue.

Classification is by layer count (first name) and cell shape (second name). Simple = one layer, made for exchange or secretion. Stratified = multiple layers, made for protection. Squamous = flat (good for diffusion). Cuboidal = cubed (room for secretion machinery). Columnar = tall (room for both, with microvilli for absorption). Once they have the naming logic, they can predict structure from function and vice versa.

Special cases: pseudostratified columnar (looks layered but is not, lines the upper airway, has cilia) and transitional (urinary bladder, can stretch flat). Spend extra time on the upper airway: pseudostratified ciliated columnar with goblet cells is the mucociliary escalator. When that fails (smoking, cystic fibrosis), the lungs lose their main defense. Predicting structure from function is the highest-yield exam skill in this section.

Teaching This Topic

Before the video (drawing prompt)

Show students a list of locations (alveolus, esophagus, kidney tubule, bladder, upper airway, small intestine) and have them predict the epithelial type for each before watching.

Common misconceptions to address

- Pseudostratified is not actually stratified. Every cell still touches the basement membrane.
- Transitional epithelium does not appear transitional under a microscope; the name means it transitions in appearance when the bladder fills.
- Students confuse 'simple' for 'simple to understand.' It means one cell layer.

Order of operations (what to teach first)

Function categories (barrier, exchange, secretion), then the naming system, then examples by location, then the special cases.

Self-test prompts

- Name the epithelium of an alveolus and explain why.
- Why is the upper airway pseudostratified with cilia rather than stratified squamous?
- What changes about transitional epithelium between an empty and full bladder?

Why This Matters to Your Patients

Chronic smokers undergo squamous metaplasia of the upper airway: protective stratified squamous replaces the original ciliated columnar. The new tissue cannot clear mucus, and pathogens linger. This is the cellular start of COPD. Burns destroy epidermis (stratified squamous) and demand grafting because the protective barrier is the entire point of skin. Barrett's esophagus replaces esophageal stratified squamous with columnar (in response to acid reflux); the new tissue is metabolically more active and more likely to become cancerous. Almost every tissue-level diagnosis in pathology is an epithelial story.

TOPIC 2 OF 3 . WEEK 2 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Connective Tissues

Cells embedded in a matrix: loose, dense, cartilage, bone, blood.

The Science

Connective tissue is defined by what is between the cells, not the cells themselves. Cells suspended in an extracellular matrix of ground substance plus protein fibers (collagen, elastic, reticular). Vary the matrix and you get loose connective tissue, dense connective tissue, cartilage, bone, or blood. One family, enormous range of function.

Loose connective (areolar, adipose, reticular) is the general-purpose packing. Dense regular (tendons, ligaments) has parallel collagen for tensile strength in one direction. Dense irregular (dermis) has interwoven collagen for strength in all directions. Cartilage has chondrocytes in lacunae within a flexible matrix; it is avascular and heals poorly. Bone has osteocytes in a rigid mineralized matrix. Blood is connective tissue with a liquid matrix (plasma).

Repair maps onto vascularity. Loose connective heals fast (rich blood supply). Cartilage heals poorly (avascular; nutrients arrive by diffusion only). Bone heals well, contrary to intuition, because it is highly vascular and has active osteoblasts ready to lay down new matrix. This is why a torn meniscus often fails to heal but a broken femur usually does.

Teaching This Topic

Before the video (drawing prompt)

Ask students to list five connective tissues and rank them by how well they heal. They have to predict before the data.

Common misconceptions to address

- Blood is connective tissue. This is counterintuitive but high-yield.
- Adipose is sometimes seen as inert padding. It is metabolically and endocrinologically active.
- Students assume cartilage and bone are similar because they are both 'hard.' Their vascular biology and repair capacity are opposite.

Order of operations (what to teach first)

What defines connective tissue (matrix), then types organized by matrix composition, then repair, organized by vascularity.

Self-test prompts

- Without notes, list five connective tissue types.
- Why does cartilage heal poorly while bone heals well?
- Why is blood classified as connective tissue?

Why This Matters to Your Patients

Osteogenesis imperfecta (defective collagen) makes bones brittle but also affects tendons, ligaments, and the sclera. Ehlers-Danlos syndromes affect collagen and produce hypermobile joints and fragile skin. A meniscal tear often requires surgery because cartilage cannot self-repair; a torn ACL has limited repair capacity for the same reason. Tendinopathy is a slow disease because the blood supply to tendons is limited. Burn patients lose dermis and require grafting because the dense irregular collagen network underneath the epidermis is the structural foundation skin needs.

TOPIC 3 OF 3 . WEEK 2 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Muscle and Nervous Tissue Overview

Three muscle types compared, plus neurons and glia at a tissue level.

The Science

Three muscle types, two nervous tissue cells: a small set, but high-yield. Compare-and-contrast pedagogy works best here.

Skeletal muscle: striated, multinucleated, voluntary, attached to bone. Cardiac: striated, single nucleus, involuntary, has intercalated discs (gap junctions that allow the heart to act as a functional syncytium). Smooth: not striated, single nucleus, involuntary, walls of hollow organs (gut, vessels, bladder, uterus). The discriminating features are striation (yes for skeletal and cardiac, no for smooth) and nucleus count (many for skeletal, one for the others).

Nervous tissue is neurons plus glia. Neurons are the excitable, signaling cells. Glia are the support cells: in the CNS, astrocytes (blood-brain barrier and metabolism), oligodendrocytes (myelin), microglia (immune), ependymal (CSF production). In the PNS, Schwann cells make myelin and satellite cells support cell bodies. Glia outnumber neurons and do most of the day-to-day maintenance. When myelin fails (multiple sclerosis in CNS, Guillain-Barre in PNS), conduction fails.

Teaching This Topic

Before the video (drawing prompt)

Have students predict, for each muscle type: striated yes or no, nuclei one or many, voluntary or not, where in the body.

Common misconceptions to address

- Cardiac muscle is sometimes lumped with skeletal because both are striated. The regulation and conduction biology are very different.
- Smooth muscle is thought of as weak. It is enormously strong; the uterus in labor and the GI tract are smooth muscle.
- Glia are thought of as background. They do most of the work of the nervous system.

Order of operations (what to teach first)

Three muscle types side-by-side (compare on a small set of features), then neurons as excitable cells, then glia as the support network.

Self-test prompts

- Which muscle type has intercalated discs and why does that matter?
- What CNS cell makes myelin?
- What goes wrong physiologically when myelin is destroyed?

Why This Matters to Your Patients

Multiple sclerosis destroys CNS myelin and produces conduction failure across whatever tract is affected: vision changes if the optic nerve is hit, weakness or numbness if spinal tracts are hit, coordination loss if cerebellar pathways are hit. Guillain-Barre syndrome attacks PNS myelin and produces ascending paralysis. Smooth muscle disorders include irritable bowel syndrome (dysregulated gut motility), urinary incontinence (detrusor dysfunction), and asthma (bronchial smooth muscle hyperreactivity). Cardiac myocyte loss is the substrate of heart failure. Three muscle types, three families of disease.

Module 4: Integumentary System

Topics: Skin Structure and Layers, Skin Functions and Accessory Structures

TOPIC 1 OF 2 . WEEK 2 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Skin Structure and Layers

Epidermis, dermis, hypodermis: who lives where and what they do.

The Science

Skin is your largest organ by surface area. Two layers strictly (epidermis, dermis) plus a hypodermis underneath that is technically not skin but is functionally part of the system. Each layer has a job.

Epidermis: stratified squamous keratinized, avascular, several sub-layers (basale, spinosum, granulosum, lucidum in thick skin only, corneum). Keratinocytes are born at the basale, mature upward, die, and form the cornified layer that protects against everything from microbes to water loss. Melanocytes in the basale produce melanin and transfer it to keratinocytes for UV protection. Langerhans cells (immune surveillance) and Merkel cells (touch) live here too.

Dermis: vascular, where the action is. Papillary layer has capillary loops and fine sensory endings; reticular layer is dense irregular connective tissue, the source of skin's strength. Hair follicles, glands, and nerves are dermal. Hypodermis is adipose and loose connective tissue, providing insulation and energy storage.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a cross-section of skin and try to label five layers/structures before watching.

Common misconceptions to address

- Students think the epidermis is vascular. It is avascular; it lives off diffusion from the dermis.
- Tanning is sometimes seen as healthy. Tanning is the keratinocyte response to UV damage; melanin is protective but the damage has already occurred.
- The hypodermis is often called part of the skin. Technically it is not, but functionally it is inseparable.

Order of operations (what to teach first)

Three layers overview, then epidermis with sub-layers, then dermis with vascular and sensory content, then hypodermis. Build from outside in.

Self-test prompts

- List the layers of the epidermis from deep to superficial in thick skin.
- Why is the epidermis avascular?
- Where in the skin would a needle for an intradermal injection actually deliver the medication?

Why This Matters to Your Patients

Burn depth: first-degree burns are epidermal (sunburn-like, painful, no scar). Second-degree are partial-thickness (blister, painful, may scar). Third-degree are full-thickness (extending into or through the dermis; the tissue from which new skin would regrow is gone, requiring grafting). Pressure ulcers, IV infiltrations, melanoma screening, transdermal drug delivery, all of these depend on knowing exactly where in the skin you are. A nurse who can name skin layers reads burns, wounds, and skin lesions with confidence.

TOPIC 2 OF 2 . WEEK 2 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Skin Functions and Accessory Structures

Protection, thermoregulation, sensation, vitamin D; plus hair, glands, and nails.

The Science

Skin functions are not a list to memorize, they are a checklist for clinical assessment. Protection (barrier, immunity), thermoregulation (sweat and vasomotor), sensation (touch, pressure, temperature, pain), vitamin D synthesis (UVB on 7-dehydrocholesterol), and minor excretion (small amounts of nitrogen waste in sweat). Lose any of these and the patient suffers.

Accessory structures: hair (insulation, sensation, identification), nails (protection of digit tips, fine manipulation), eccrine sweat glands (thermoregulation), apocrine sweat glands (axilla and groin, activated at puberty, produce odor when bacteria feed on the secretion), sebaceous glands (oily sebum, waterproofs, slightly antimicrobial). Each accessory structure is a specialization of skin tissue.

Thermoregulation deserves a beat: vasodilation brings warm blood to the skin surface for heat dissipation (radiation, conduction), sweating cools by evaporation (water's high heat of vaporization, see Chemistry of Life), vasoconstriction conserves core heat. The countercurrent arrangement between cutaneous arteries and veins is an additional layer of thermal control.

Teaching This Topic

Before the video (drawing prompt)

Ask students to list every function of skin they can think of in 60 seconds, then compare to your list. Most will get fewer than half.

Common misconceptions to address

- Skin is seen as cosmetic. It is the body's largest immune and thermoregulatory organ.
- Sweat is thought to cool by 'feeling cool.' It cools by evaporating; if it just drips off without evaporating, no cooling occurs.
- Vitamin D from skin is often overlooked. Sun exposure is a major route; deficiency is common in dark-skinned people at high latitudes.

Order of operations (what to teach first)

Functions first (with thermoregulation expanded), then accessory structures and how each contributes.

Self-test prompts

- Name five functions of the integumentary system.
- Explain how sweating actually cools the body.
- Why are dark-skinned people at higher risk of vitamin D deficiency in northern climates?

Why This Matters to Your Patients

Burn patients lose fluid through damaged skin at staggering rates and die of dehydration, hypothermia, and infection long before the burn itself would kill them. Diabetics with poor circulation develop pressure ulcers because skin cannot defend or repair itself without good blood flow. Patients with autonomic dysfunction lose thermoregulation; spinal cord injury patients above T6 cannot vasoconstrict below the level of injury and are at risk for hypothermia in mild cold. Vitamin D status affects bone health, immunity, and probably mood; supplementation matters. Skin assessment is the most accessible window into systemic disease the nurse has.

Module 5: Skeletal System

Topics: Bone Tissue and Bone Growth, Axial Skeleton, Appendicular Skeleton, Joints and Body Movements

TOPIC 1 OF 4 . WEEK 3 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Bone Tissue and Bone Growth

Cells of bone, compact vs spongy, intramembranous vs endochondral ossification, and lifelong remodeling.

The Science

Bone is connective tissue with a mineralized matrix. It is alive and constantly being remodeled. The cells are osteoblasts (build matrix), osteocytes (mature, trapped in lacunae, sense load), and osteoclasts (resorb matrix; derived from monocyte lineage, not osteoblasts). Osteoblast versus osteoclast balance determines whether bone is being added or removed.

Compact bone is dense, organized into osteons (Haversian systems). Each osteon has concentric lamellae of matrix around a central canal carrying vessels and nerves. Spongy bone is the trabecular meshwork in the interior of bones, often containing red marrow. Long bones have a diaphysis (shaft), epiphyses (ends), metaphysis (growth zone), periosteum (outer covering), and endosteum (lining the marrow cavity).

Two kinds of ossification: intramembranous (skull flat bones, clavicle) directly from mesenchyme, and endochondral (most other bones) from a hyaline cartilage model. Lengthening occurs at the epiphyseal plate until it closes in late adolescence. Throughout life, remodeling continues. PTH raises blood calcium by activating osteoclasts; calcitonin opposes it (modest effect). Estrogen restrains osteoclasts; this is why postmenopausal women lose bone rapidly.

Teaching This Topic

Before the video (drawing prompt)

Ask students to draw a long bone in cross-section and label five structures. They have to commit before checking.

Common misconceptions to address

- Bone is thought of as inert. Bone is intensely metabolically active and continuously remodeled.
- Osteoclast and osteoblast names sound similar; students mix them up. Cue: clast like clast-rophobic = breaking down; blast like blast off = building.
- Calcium balance and bone density are sometimes seen as the same thing. They are linked but distinct: short-term calcium balance can be at the expense of bone density.

Order of operations (what to teach first)

Cells, then compact vs spongy structure, then ossification types, then lifelong remodeling and hormonal control.

Self-test prompts

- Cue yourself: what does each bone cell do?
- How does estrogen affect bone density and why does postmenopausal osteoporosis happen?
- What is happening at the epiphyseal plate and why does it close in adulthood?

Why This Matters to Your Patients

Osteoporosis is bone loss from accelerated osteoclast activity relative to osteoblasts. Postmenopausal women are at high risk because estrogen normally restrains osteoclasts. Bisphosphonates are first-line treatment; they inhibit osteoclasts. Hyperparathyroidism raises blood calcium by mobilizing bone calcium, producing kidney stones, fractures, and 'stones, bones, groans' on the boards. Pediatric fracture management depends on whether the epiphyseal plate is involved (Salter-Harris classification); plate injuries can disrupt growth. Bone metastases are common in advanced cancers and cause pain, fractures, and hypercalcemia. Nurses caring for elderly patients are constantly thinking about fall risk, fracture prevention, and bone density.

TOPIC 2 OF 4 . WEEK 3 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Axial Skeleton

Skull, vertebral column, and thoracic cage: the central axis of the body.

The Science

The axial skeleton is the central line: skull, vertebral column, and thoracic cage. Its job is protection of the most vulnerable organs (brain, spinal cord, heart, lungs) and anchorage for the appendicular skeleton.

Skull: eight cranial bones (frontal, parietal pair, temporal pair, occipital, sphenoid, ethmoid) and fourteen facial bones (maxillae, zygomatics, mandible, and the smaller facial bones). Sutures are immovable joints. The vertebral column has five regions: cervical (7, including the atlas and axis), thoracic (12, articulating with ribs), lumbar (5, the weight-bearers), sacrum (5 fused), coccyx (3-4 fused).

Thoracic cage: sternum (manubrium, body, xiphoid) and 12 pairs of ribs (true 1-7, false 8-10, floating 11-12). Function: protect heart and lungs, anchor breathing muscles. The shape and mobility of the cage are critical for ventilation; chronic obstructive lung disease patients develop a barrel chest because their inflated lungs remodel the cage.

Teaching This Topic

Before the video (drawing prompt)

Ask students to list the five regions of the vertebral column and the count of vertebrae in each, from memory.

Common misconceptions to address

- Students think all vertebrae look the same. Each region has distinctive shape because of different function: cervical for mobility, thoracic for rib articulation, lumbar for weight-bearing.
- Floating ribs are thought of as defects. They are normal anatomy.
- The sacrum is treated as a single bone; it is five fused vertebrae and the fusion is functionally important.

Order of operations (what to teach first)

Skull, then vertebral column with regional contrasts, then thoracic cage and its ventilatory role.

Self-test prompts

- Without notes, write the number of vertebrae in each region.
- What is unique about C1 (atlas) and C2 (axis) and why?
- How does a compression fracture of a thoracic vertebra change posture and breathing?

Why This Matters to Your Patients

Compression fractures of thoracic and lumbar vertebrae are common in osteoporosis and produce loss of height, kyphosis ('dowager's hump'), and chronic pain. Rib fractures limit ventilation because breathing hurts; nurses watch for atelectasis and pneumonia. Skull fractures depend on where they are: a basilar skull fracture can damage cranial nerves; a temporal bone fracture can rupture the middle meningeal artery and cause an epidural hematoma. Spinal cord injuries are classified by level: a C5 transection preserves diaphragm function but paralyzes the arms and legs; C3 or above paralyzes the diaphragm and the patient needs ventilator support.

TOPIC 3 OF 4 . WEEK 3 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Appendicular Skeleton

Pectoral girdle and upper limb, pelvic girdle and lower limb.

The Science

The appendicular skeleton is everything that hangs off the axial. Pectoral girdle (clavicle and scapula) attaches the upper limb. Pelvic girdle (two coxal bones plus the sacrum) attaches the lower limb. The trade-off across the two girdles is instructive: pectoral is loose (mobility), pelvic is fused (stability).

Upper limb: humerus (arm), radius (lateral, thumb side) and ulna (medial) (forearm), 8 carpals (wrist), 5 metacarpals (palm), 14 phalanges (fingers). Lower limb: femur (thigh, largest bone in the body), tibia (medial weight-bearer) and fibula (lateral) (leg), 7 tarsals (ankle and back of foot), 5 metatarsals (foot), 14 phalanges (toes).

Pelvic girdle has key sex differences: female pelvis is wider, shallower, with a larger pelvic inlet, designed for childbirth. The pubic angle is wider in females (about 100 degrees) versus narrower in males (about 60 degrees). These are clinically relevant in obstetrics, pelvic surgery, and forensic anthropology.

Teaching This Topic

Before the video (drawing prompt)

Ask students to draw the bones of one arm and one leg from memory, labeling. They have to commit before correction.

Common misconceptions to address

- Students confuse radius and ulna. Cue: radius is on the same side as the radial artery (thumb side).
- Tibia and fibula get mixed up. Tibia is the weight-bearer (much thicker); fibula is the lateral splint.
- Students assume male and female skeletons are identical. The pelvis differs significantly.

Order of operations (what to teach first)

Pectoral girdle, upper limb proximal to distal, then pelvic girdle, then lower limb. Compare girdle mobility versus stability as a closing concept.

Self-test prompts

- Which forearm bone is on the thumb side?
- How many bones in the wrist?
- How does the female pelvis differ from the male and why?

Why This Matters to Your Patients

Clavicle fractures are common (the most common fracture in children) and usually heal without surgery, but the shoulder drops because the strut is broken. Hip fractures in the elderly are a major source of morbidity; mortality at one year is around 20-30%. Carpal tunnel syndrome (median nerve compression at the wrist) is a workplace injury epidemic. Pelvic fractures from high-energy trauma can hide enormous blood loss in the pelvic cavity. Obstetric nursing depends on knowing pelvic anatomy in detail.

TOPIC 4 OF 4 . WEEK 3 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Joints and Body Movements

Classifying joints by structure and function, plus the vocabulary of movement.

The Science

Joints classify two ways. Structurally: fibrous, cartilaginous, synovial, based on what connects the bones. Functionally: synarthrosis (immovable), amphiarthrosis (slightly movable), diarthrosis (freely movable). Most synovial joints are diarthroses; most fibrous and many cartilaginous joints are synarthroses or amphiarthroses.

Synovial joints have the most clinical relevance. They have a joint cavity filled with synovial fluid, articular cartilage on bone ends, a fibrous capsule, and often ligaments, menisci, and bursae. Subtypes: plane (carpals), hinge (elbow, knee, interphalangeal), pivot (atlanto-axial), condyloid (wrist), saddle (thumb), ball-and-socket (shoulder, hip). Each subtype has characteristic movements.

Movements: flexion (decreases joint angle), extension (increases it), abduction (moves away from midline), adduction (moves toward midline), rotation (around long axis), circumduction (cone-shaped sweep), plus the specialty terms (pronation, supination, inversion, eversion, dorsiflexion, plantarflexion). Teach the major terms with both the verbal definition and a body demonstration.

Teaching This Topic**Before the video (drawing prompt)**

Have students perform each movement (flexion, extension, abduction, etc.) on themselves and write the joint that allows it. Movement memory is stronger than verbal memory.

Common misconceptions to address

- Students confuse pronation and supination. Cue: supination = palm up, like carrying SOUP.
- Flexion is sometimes thought to mean 'curling.' It means decreasing the joint angle, which is curling for some joints but not for others (the shoulder flexes by raising the arm forward).
- Hinge joints are thought to be limited to one axis. They are, but with some rotation in the knee (which is unusual).

Order of operations (what to teach first)

Classification (structural and functional), then synovial subtypes with examples, then movements with physical demonstration.

Self-test prompts

- Name the six synovial joint subtypes and give an example of each.
- What is the difference between abduction and adduction?
- Demonstrate flexion and extension of the elbow, hip, and shoulder.

Why This Matters to Your Patients

Rheumatoid arthritis attacks synovial joints; the inflamed synovium forms a pannus that erodes cartilage, then bone, leaving deformed joints. Osteoarthritis is wear and tear: cartilage thins, joint space narrows, osteophytes form. Joint dislocations are more common at the shoulder than the hip because the shoulder trades stability for mobility. Ligament injuries (ACL, MCL of the knee) are common in athletes. Hip replacements and knee replacements are among the most common surgeries in the elderly population. Knowing joint anatomy is part of every assessment of a patient who fell, had a sports injury, or has chronic joint pain.

Module 6: Muscular System

Topics: Skeletal Muscle Microanatomy, Sliding Filament and the Cross-Bridge Cycle, Motor Units and Muscle Mechanics

TOPIC 1 OF 3 . WEEK 4 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Skeletal Muscle Microanatomy

From whole muscle down to the sarcomere: the structural ladder of contractile tissue.

The Science

The structural hierarchy is the spine of the topic: muscle, fascicle, fiber (cell), myofibril, sarcomere. Each level is wrapped in connective tissue (epimysium, perimysium, endomysium) that converges into the tendon. Build this scaffold first; everything else hangs on it.

The sarcomere is the contractile unit. Z-line to Z-line. Thick filaments (myosin) in the center (A band). Thin filaments (actin, troponin, tropomyosin) extending from the Z-line (I band). The overlap region is where cross-bridges can form. During contraction the I band shrinks and the H zone (myosin only) narrows, but the A band (thick filament length) does not change. This is a common test point and a useful conceptual anchor.

Excitation machinery: sarcolemma (the muscle fiber's plasma membrane), T-tubules (deep invaginations that carry action potentials into the interior), sarcoplasmic reticulum (SR, the calcium store wrapping each myofibril). T-tubules and SR meet at triads, where electrical excitation triggers calcium release. This coupling is essential; disrupt the triads and contraction fails even though action potentials still propagate.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the structural hierarchy as nested levels, then a sarcomere with bands labeled. Pre-commitment makes the next topic (sliding filament) much easier.

Common misconceptions to address

- Students think the A band changes length during contraction. It does not; the thick filament is rigid.
- T-tubules and SR are confused. T-tubules carry action potentials inward; SR stores and releases calcium. Both meet at the triad.
- Sarcomere is sometimes called sarcomeres (plural). Get the singular cue: it is one unit, repeated.

Order of operations (what to teach first)

Hierarchy first, then sarcomere structure, then excitation machinery. Save the cross-bridge cycle for the next topic; it is its own beat.

Self-test prompts

- Without notes, list the structural levels from whole muscle down to sarcomere.
- Sketch a sarcomere and label A band, I band, H zone, Z-line.
- What is the function of a triad?

Why This Matters to Your Patients

Muscular dystrophies are diseases of structural proteins (dystrophin in Duchenne). The sarcolemma cannot withstand the mechanical stress of contraction, fibers break down, and progressive weakness results. Malignant hyperthermia (next topic) involves a defective SR calcium channel. Statins occasionally cause myopathy: muscle pain and weakness from sarcomere-level disturbance. Understanding skeletal muscle microanatomy is understanding why a patient with myopathy is weak and what tests confirm it (CK elevation, biopsy).

TOPIC 2 OF 3 . WEEK 4 . 10 CARDS (5 DOK 1 / 3 DOK 2 / 2 DOK 3)

Sliding Filament and the Cross-Bridge Cycle

How calcium, ATP, actin, and myosin convert chemical energy into mechanical force.

The Science

This is one of the most beautiful pieces of physiology and one of the highest-yield topics in A&P. Spend the time.

Step 0 (the trigger): action potential at the neuromuscular junction releases ACh, opens nicotinic receptors, depolarizes the sarcolemma. The depolarization sweeps down T-tubules and triggers calcium release from the SR. Calcium is the signal.

The four-step cross-bridge cycle: (1) Cocking, myosin head hydrolyzes ATP to ADP+Pi and reorients into the high-energy position. (2) Binding, calcium binds troponin, tropomyosin shifts off the actin sites, myosin binds actin. (3) Power stroke, myosin pivots, pulling actin toward the M-line, releasing ADP and Pi. (4) Detachment, a new ATP binds myosin, which releases actin; cycle repeats. The thin filaments slide along the thick filaments; the sarcomere shortens; the muscle contracts. Filament lengths do not change; their overlap

does.

Relaxation: when neural input stops, ACh is broken down by acetylcholinesterase, the sarcolemma repolarizes, the SR calcium ATPase pumps calcium back into the SR, troponin returns to its resting position, tropomyosin re-covers the actin sites. No calcium, no cross-bridges, no contraction. Both excitation and relaxation require ATP. This is why rigor mortis happens: after death, no ATP means myosin cannot detach from actin.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a sarcomere at rest and predict what changes during contraction. Then they have to draw the four-step cycle in order before watching.

Common misconceptions to address

- Students think calcium directly attaches myosin to actin. Calcium binds troponin; troponin moves tropomyosin; tropomyosin exposes the actin site.
- ATP is thought to be needed only for the power stroke. It is needed for cocking AND detachment, but the power stroke itself is the release of stored energy.
- Rigor mortis is sometimes thought to be from low calcium. It is from no ATP, so no detachment, so locked cross-bridges.

Order of operations (what to teach first)

Trigger first (ACh to calcium), then the four steps drilled IN ORDER with hand gestures, then relaxation, then rigor as a check on understanding.

Self-test prompts

- Without notes, list the four steps of the cross-bridge cycle in order.
- Where in the cycle is ATP hydrolyzed? Where is fresh ATP needed?
- Explain rigor mortis in terms of cross-bridge biology.

Why This Matters to Your Patients

Curare and similar drugs block nicotinic ACh receptors at the neuromuscular junction, preventing depolarization, causing flaccid paralysis. Used clinically for surgical muscle relaxation. Organophosphate poisoning inhibits acetylcholinesterase, so ACh lingers, producing fasciculations and eventually depolarization block and respiratory paralysis (the farm worker's emergency, the sarin gas attack). Malignant hyperthermia is a genetic defect in the SR calcium release channel: certain anesthetics trigger uncontrolled calcium release, sustained contraction, hyperthermia from continuous cross-bridge cycling, ATP depletion, and rhabdomyolysis. Treatment is dantrolene (blocks SR calcium release). Every one of these stories is one node of the cross-bridge cycle being attacked.

TOPIC 3 OF 3 . WEEK 4 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Motor Units and Muscle Mechanics

Motor unit organization, recruitment, summation, and fatigue.

The Science

A motor unit is one motor neuron plus all the muscle fibers it innervates. Small motor units (a few fibers) give fine control: extraocular muscles have motor units of 5-10 fibers. Large motor units (many fibers) give power: quadriceps motor units have thousands. The size of the motor unit is set by what the muscle needs to do.

Force grading happens two ways. Recruitment: activate more motor units (the Henneman size principle says small units fire first, large units recruited as force demand rises). Frequency summation: fire the same motor unit faster, so twitches overlap. If firing is fast enough, twitches fuse into a sustained, smooth contraction (tetanus). These two mechanisms are how the body smoothly grades force from picking up a feather to deadlifting a barbell.

Fiber types: Type I (slow oxidative, fatigue-resistant, postural muscles), Type IIa (fast oxidative, intermediate), Type IIx (fast glycolytic, powerful but fatigue quickly). Endurance training shifts the population toward Type I characteristics; power training toward Type IIx. Fatigue has multiple causes: substrate depletion (glycogen, ATP), metabolic byproduct accumulation (H^+ , P_i), and neural factors (central fatigue).

Teaching This Topic

Before the video (drawing prompt)

Ask students to predict the motor unit size in an eye muscle versus a thigh muscle. They have to commit before the answer.

Common misconceptions to address

- Tetanus the disease is named after tetanus the contraction, which causes the rigid muscles. Students often miss this connection.
- Students think 'recruitment' is about overall fitness. It is the moment-to-moment neural decision to activate more motor units.
- Fast and slow twitch fibers are sometimes thought to be fixed at birth. They are largely genetic but training shifts the population.

Order of operations (what to teach first)

Motor unit definition, then force grading (recruitment and frequency summation), then fiber types and adaptation.

Self-test prompts

- Why are extraocular muscles innervated by very small motor units?
- What is the difference between recruitment and frequency summation?
- Predict the fiber type adaptations in an endurance athlete versus a sprinter.

Why This Matters to Your Patients

Myasthenia gravis is an autoimmune attack on nicotinic ACh receptors at the NMJ, producing fluctuating weakness that worsens with repeated use. Lambert-Eaton syndrome is an attack on the presynaptic voltage-gated calcium channel; the resulting decrease in ACh release produces weakness that improves with repeated activation (opposite of myasthenia). Critical illness myopathy and disuse atrophy occur in ICU patients within days of immobilization. ALS destroys motor neurons themselves, denervating muscles and producing progressive paralysis. Nurses caring for any patient with weakness need to think about what part of the motor system is failing.

Module 7: Nervous System

Topics: Neurons and Resting Membrane Potential, Action Potentials and Synaptic Transmission, CNS Organization: Brain and Spinal Cord, PNS and Autonomic Nervous System

TOPIC 1 OF 4 . WEEK 4 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Neurons and Resting Membrane Potential

Neuron anatomy and how the resting potential is built and maintained.

The Science

Neuron anatomy first: dendrites receive, soma integrates, axon transmits, axon terminal releases neurotransmitter. The neuron is shaped for its job: a polarized communication device that can be a few millimeters long (interneuron) or over a meter long (motor neuron from spinal cord to foot).

Resting membrane potential is about -70 mV: the inside of the cell is negative relative to the outside. Two main forces keep it that way: the Na/K ATPase actively pumps 3 Na out and 2 K in (net export of positive charge, slightly negative contribution to the potential), and K leak channels let K flow out down its concentration gradient (leaving the inside more negative). The membrane at rest is mostly permeable to K; this is why the resting potential sits close to the K equilibrium potential of -90 mV but not exactly there (small contributions from Na and Cl).

Glia matter: astrocytes maintain the blood-brain barrier and clean up extracellular K and neurotransmitter. Oligodendrocytes wrap axons in myelin (CNS); Schwann cells do the same job in the PNS. Microglia are the immune cells of the brain. Ependymal cells line the ventricles and produce CSF. Glia outnumber neurons by roughly 10:1; the nervous system is mostly glia, not neurons.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a neuron with the parts labeled and then a graph of membrane potential over time at rest. They have to commit to the resting value.

Common misconceptions to address

- Students think the resting potential is set entirely by the Na/K pump. The pump is necessary to build the gradients, but the resting potential is set by ion permeability (mostly K leak).
- The minus sign of -70 mV is sometimes ignored. It is the entire point: the cell interior is negative.
- Glia are seen as bystanders. They are essential and outnumber neurons.

Order of operations (what to teach first)

Anatomy first, then the gradients (Na out, K in), then permeability and the resting potential, then glia as context.

Self-test prompts

- What is the typical resting membrane potential of a neuron?
- Why is the inside of the cell negative?
- Name the CNS glial cells and what each one does.

Why This Matters to Your Patients

Local anesthetics block voltage-gated Na channels and prevent action potentials in sensory neurons; the patient loses pain sensation. Cardiac glycosides (digoxin) inhibit the Na/K ATPase, which raises intracellular Na, which in turn raises intracellular Ca (through the Na/Ca exchanger), strengthening cardiac contraction. Multiple sclerosis (demyelination of CNS axons) is a glial disease. Hypokalemia and hyperkalemia both produce dangerous changes in membrane excitability, which is why potassium is one of the most carefully monitored electrolytes in the hospital. Every nurse needs to understand the relationship between potassium and the resting potential.

TOPIC 2 OF 4 . WEEK 4 . 12 CARDS (5 DOK 1 / 3 DOK 2 / 4 DOK 3)

Action Potentials and Synaptic Transmission

Phases of an action potential, propagation, and how chemical synapses pass the signal on.

The Science

The action potential is one of physiology's signature stories. Spend time on it; this is high-yield.

Phases: at rest, -70 mV. A stimulus depolarizes the membrane; if it reaches threshold (about -55 mV), voltage-gated Na channels open en masse, Na rushes in down its gradient, and the membrane swings positive (to about +30 mV). Then Na channels inactivate (a separate

gate closes them) and voltage-gated K channels open; K rushes out, repolarizing the cell. K channels close slowly, producing a brief afterhyperpolarization. The Na/K pump resets the gradients (slowly, in the background). All-or-none: a suprathreshold stimulus always produces the same-size AP.

Propagation: unmyelinated axons propagate continuously, regenerating the AP at every patch of membrane (slow). Myelinated axons regenerate the AP only at nodes of Ranvier; the signal effectively jumps from node to node (saltatory conduction, much faster).

Refractory periods (absolute, then relative) prevent backward propagation and limit firing frequency.

Synaptic transmission: AP reaches the terminal, opens voltage-gated Ca channels, Ca rushes in, vesicles fuse with the membrane, neurotransmitter is released into the synaptic cleft, binds receptors on the postsynaptic membrane, produces an EPSP (excitatory) or IPSP (inhibitory). Signal termination: reuptake, enzymatic breakdown, or diffusion away. EPSPs and IPSPs summate at the postsynaptic axon hillock; if the net depolarization reaches threshold, the postsynaptic cell fires.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a labeled action potential trace from memory: time on x-axis, voltage on y-axis, with the phases marked. They have to commit before the video.

Common misconceptions to address

- Bigger stimulus equals bigger action potential. False: all-or-none means the AP is the same size regardless of stimulus strength. Bigger stimulus equals more frequent APs.
- K and Na are confused as to direction. Cue: Na is in for depolarization, K is out for repolarization.
- Saltatory conduction is sometimes described as 'jumping' which makes students think the AP literally jumps over membrane. It is regenerated at each node, just not at intervening membrane.

Order of operations (what to teach first)

Resting potential recap, then the AP phases (in order with ion movements), then all-or-none and refractory periods, then propagation, then synaptic transmission as the next-cell handoff.

Self-test prompts

- From memory, label the phases of an action potential on a voltage trace.
- Explain saltatory conduction in one sentence.
- What ion triggers neurotransmitter release at the axon terminal?

Why This Matters to Your Patients

Multiple sclerosis demyelinates CNS axons; saltatory conduction fails; symptoms depend on which tracts are affected. Local anesthetics block Na channels in sensory neurons selectively (because sensory axons are smaller and more easily blocked) and produce regional anesthesia. SSRIs raise synaptic serotonin by blocking reuptake; the antidepressant effect lags weeks behind the immediate pharmacology because downstream neuroplasticity changes are slow. Botulinum toxin blocks neurotransmitter release (presynaptic) and causes flaccid paralysis; tetanus toxin blocks inhibitory neurotransmitter release and causes rigid paralysis. Almost every neurology drug acts on some node of the AP-synapse cascade.

TOPIC 3 OF 4 . WEEK 4 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

CNS Organization: Brain and Spinal Cord

Brain regions and lobes, meninges, ventricles, and spinal cord anatomy.

The Science

The brain has four major regions to get right. Cerebrum: the cortex (gray matter) over white matter, divided into four lobes (frontal, parietal, temporal, occipital). Diencephalon: thalamus (sensory relay), hypothalamus (homeostasis HQ), epithalamus (pineal gland). Brainstem: midbrain, pons, medulla (vital reflexes, cranial nerves III-XII). Cerebellum: motor coordination and balance.

Functional cortex landmarks: precentral gyrus (frontal lobe) is the primary motor cortex; postcentral gyrus (parietal lobe) is the primary somatosensory cortex. Occipital lobe has the primary visual cortex; temporal lobe has auditory cortex and Wernicke area (language comprehension); Broca area is in the frontal lobe (language production). These landmarks are how you read a stroke.

Coverings and fluid: three meningeal layers (dura outside, arachnoid, pia next to the brain). CSF is made by choroid plexus in the ventricles, circulates through the subarachnoid space, and drains via arachnoid villi into the venous system. The spinal cord runs from the medulla to L1-L2. Below L1-L2, the cauda equina (loose bundle of spinal nerve roots) is what you stick a needle around for a lumbar puncture.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the cerebrum, label the four lobes, and put a star on the precentral gyrus, postcentral gyrus, Broca area, and Wernicke area. They have to commit before the video.

Common misconceptions to address

- Students think the cerebellum makes movements happen. It modulates and coordinates movements; the motor cortex initiates them.
- The brainstem is often dismissed as small. It contains the vital reflex centers for breathing, heart rate, and consciousness.
- Broca and Wernicke get confused. Cue: Broca (frontal) is for production ('mouth in front'), Wernicke (temporal) is for comprehension ('ears on side').

Order of operations (what to teach first)

Four major regions, then cortex landmarks, then meninges and CSF, then spinal cord as the extension into the body.

Self-test prompts

- Name the four lobes of the cerebrum.
- Where is the primary motor cortex?
- What is the difference between Broca aphasia and Wernicke aphasia?

Why This Matters to Your Patients

Strokes are localized by neurological deficit. A patient with right-sided weakness, right-sided sensory loss, and Broca-type aphasia has had a left middle cerebral artery stroke. A patient with vertical gaze palsy, ipsilateral facial weakness, and contralateral body weakness has a brainstem lesion. Cerebellar strokes produce ataxia (uncoordinated movement) without paralysis. Increased intracranial pressure produces a stereotyped triad: hypertension, bradycardia, irregular breathing (Cushing triad). Nurses are often the first to spot neurological changes by serial assessment; knowing the anatomy means knowing where the problem is.

TOPIC 4 OF 4 . WEEK 4 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

PNS and Autonomic Nervous System

Cranial and spinal nerves, reflex arcs, and the sympathetic vs parasympathetic divisions.

The Science

PNS = cranial nerves and spinal nerves. Twelve pairs of cranial nerves (mostly head and neck; vagus, CN X, is the long-haul exception, innervating the heart, lungs, and much of the GI tract). Thirty-one pairs of spinal nerves, each with sensory (dorsal root) and motor (ventral root) fibers.

Somatic motor vs autonomic motor. Somatic is voluntary, single neuron from CNS to skeletal muscle, releases ACh at the NMJ. Autonomic is involuntary, two-neuron chain (pre-ganglionic from CNS, synapses in a ganglion, post-ganglionic to target), innervates smooth muscle, cardiac muscle, and glands.

Sympathetic ('fight or flight'): thoracolumbar origin (T1-L2). Short pre-ganglionic, long post-ganglionic. Most post-ganglionic releases norepinephrine at the target. Effects: pupils dilate, heart rate and force up, bronchi dilate, gut motility down, skeletal muscle vasodilation, skin vasoconstriction, glycogen breakdown. Designed for action.

Parasympathetic ('rest and digest'): craniosacral origin (CN III, VII, IX, X plus S2-S4). Long pre-ganglionic, short post-ganglionic. Releases ACh at target. Effects: pupils constrict, heart rate down, bronchi constrict, gut motility up, digestion active. Designed for maintenance.

Most organs receive both; the balance shifts depending on state. The reflex arc is the basic functional unit: receptor, sensory neuron, integration center (often spinal cord), motor neuron, effector.

Teaching This Topic

Before the video (drawing prompt)

Ask students to predict the sympathetic effect on each of: pupil, heart rate, gut, bronchi, glycogen. Then test the same list under parasympathetic activation.

Common misconceptions to address

- Sympathetic and parasympathetic are seen as opposing forces in a tug-of-war. More accurately, they have different baselines for different organs, and the balance shifts.

- Both divisions release ACh at some point. Cue: parasympathetic uses ACh at the target; sympathetic uses NE at the target (with ganglia using ACh in both cases).
- Students think autonomic effects are slow because they are involuntary. Sympathetic responses can be very fast (adrenaline surge in seconds).

Order of operations (what to teach first)

PNS overview (cranial and spinal nerves), then somatic vs autonomic distinction, then sympathetic and parasympathetic in detail, then the reflex arc.

Self-test prompts

- List the five components of a reflex arc.
- What does the sympathetic system do to the eye? The heart? The gut?
- Which cranial nerve has the broadest parasympathetic distribution?

Why This Matters to Your Patients

Beta-blockers blunt sympathetic effects on the heart (slower rate, less force, lower blood pressure). Atropine blocks parasympathetic effects at muscarinic receptors and is used to raise heart rate or dry secretions. Spinal cord injuries above T6 disrupt sympathetic outflow and produce autonomic dysreflexia: a noxious stimulus below the lesion (full bladder, pressure ulcer) triggers an unopposed sympathetic surge with severe hypertension and headache, while the parasympathetic system tries to compensate above the lesion. This is a true emergency. Diabetic autonomic neuropathy produces orthostatic hypotension, gastroparesis, and erectile dysfunction. Knowing the autonomic system is knowing why many medications produce predictable side effects.

Module 8: Special Senses

Topics: Vision, Hearing and Equilibrium

TOPIC 1 OF 2 . WEEK 5 . 6 CARDS (3 DOK 1 / 2 DOK 2 / 1 DOK 3)

Vision

Eye anatomy, accommodation, and photoreceptor function.

The Science

Light hits the cornea first; the cornea does most of the eye's fixed focusing. Then the lens, which is adjustable (accommodation). Then the vitreous humor; then the retina, which contains photoreceptors.

Accommodation: for near vision the ciliary muscle contracts, the zonular fibers slacken, and the lens elastically rounds up (stronger refraction). For far vision the ciliary muscle relaxes, zonules pull the lens flatter. With age the lens stiffens (presbyopia) and near vision worsens; reading glasses compensate.

Photoreceptors: rods are highly sensitive, give black-and-white vision, dominate at low light. Cones come in three types (red, green, blue), are less sensitive but give color and acuity, dominate in daylight. The fovea is the area of sharpest vision: all cones, no overlying neurons. The optic disc is where the optic nerve leaves the eye; no photoreceptors, hence the blind spot.

Phototransduction: light isomerizes the chromophore in rhodopsin (or cone opsins), which triggers a signaling cascade that hyperpolarizes the photoreceptor. Photoreceptors release less glutamate in light. Downstream bipolar and ganglion cells interpret the change. The signal exits via the optic nerve, partially crosses at the optic chiasm, and reaches the lateral geniculate nucleus and visual cortex.

Teaching This Topic

Before the video (drawing prompt)

Have students draw an eye in cross-section and label cornea, lens, retina, fovea, optic disc, optic nerve.

Common misconceptions to address

- The lens does most of the focusing. False: the cornea does most of it; the lens fine-tunes via accommodation.
- Rods give color vision. False: rods are achromatic; cones give color.
- Students think the blind spot is at the center of vision. It is offset; you do not notice it because the brain fills in the missing region.

Order of operations (what to teach first)

Light path through the eye, then accommodation, then photoreceptor types and distribution, then the basics of phototransduction.

Self-test prompts

- What structure does most of the eye's fixed refraction?
- Why is night vision mostly black and white?
- Why is there a blind spot and where is it?

Why This Matters to Your Patients

Cataracts cloud the lens (often from oxidative stress with age); surgery replaces the lens with a synthetic one. Glaucoma raises intraocular pressure (failure of aqueous humor drainage) and damages the optic nerve; nurses screen for this in older patients. Macular degeneration destroys the fovea, robbing central vision but sparing peripheral. Diabetic retinopathy damages retinal vessels and is a leading cause of blindness. Color blindness is usually an X-linked recessive disorder of cone opsins (mostly affecting males). Visual symptoms in any patient should prompt thinking about which anatomical level is failing.

TOPIC 2 OF 2 . WEEK 5 . 5 CARDS (3 DOK 1 / 1 DOK 2 / 1 DOK 3)

Hearing and Equilibrium

Cochlear transduction of sound and vestibular detection of head motion.

The Science

Sound path: pinna and external auditory canal funnel sound to the tympanic membrane. Vibration of the tympanic membrane moves the malleus, then the incus, then the stapes. The stapes pushes on the oval window, sending pressure waves into the perilymph of the cochlea.

In the cochlea, the basilar membrane vibrates, with different frequencies producing maximum vibration at different positions (tonotopy: high frequencies at the base, low frequencies at the apex). Hair cells in the organ of Corti have stereocilia that bend with basilar

membrane motion. Bending opens mechanically gated K channels; the hair cell depolarizes; it releases glutamate onto CN VIII (the vestibulocochlear nerve), which carries the signal to the auditory cortex.

Equilibrium: the vestibule (utricle and saccule) detects linear acceleration and head tilt via otolith organs. The three semicircular canals detect rotational acceleration. All use hair cells with stereocilia in gel; head motion shears the gel and bends the cilia. Vestibular signals travel with auditory signals on CN VIII to the brainstem and cerebellum.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the ear in cross-section: outer, middle, inner; tympanic membrane, ossicles, cochlea, semicircular canals. Then label them.

Common misconceptions to address

- Students think hearing happens in the cochlea entirely. It begins at the tympanic membrane and ends in the auditory cortex.
- All hearing loss is the same. Conductive loss (middle ear) is different from sensorineural loss (cochlea or nerve).
- Vertigo from inner ear problems is sometimes confused with light-headedness from low blood pressure. Vertigo is true spinning; light-headedness is not.

Order of operations (what to teach first)

Sound path mechanically, then cochlear transduction (tonotopy), then equilibrium as a parallel hair-cell system.

Self-test prompts

- Name the three middle ear ossicles in order.
- Where in the cochlea are high frequencies detected?
- Which structures detect rotational versus linear acceleration?

Why This Matters to Your Patients

Otitis media (middle ear infection) is the most common reason a child sees a doctor; fluid behind the tympanic membrane produces conductive hearing loss. Presbycusis (age-related sensorineural hearing loss) starts with high frequencies (hair cells at the cochlear base) and progresses. Benign paroxysmal positional vertigo (BPPV) is loose otoconia drifting into a semicircular canal; head movements trigger spinning vertigo. Meniere disease is endolymph dysregulation in the inner ear, producing episodic vertigo, fluctuating hearing loss, and tinnitus. Aminoglycoside antibiotics (gentamicin) are ototoxic; nurses monitor hearing carefully during long courses.

Module 9: Endocrine System

Topics: *Hormone Mechanisms, Major Endocrine Glands*

TOPIC 1 OF 2 . WEEK 5 . 5 CARDS (3 DOK 1 / 1 DOK 2 / 1 DOK 3)

Hormone Mechanisms

Steroid vs peptide vs amino-acid-derived hormones, and how each transduces a signal.

The Science

Hormones are signaling molecules released into the blood by endocrine glands. Three chemical classes, two mechanisms.

Lipid-soluble hormones (steroids from cholesterol: cortisol, aldosterone, sex steroids; thyroid hormone): cross the plasma membrane, bind intracellular receptors, act as transcription factors. Effects are slow (hours to days) and long-lasting because they change which proteins the cell is making. Water-soluble hormones (peptides like insulin, growth hormone, ADH; catecholamines like epinephrine): cannot cross the membrane, bind cell-surface receptors, trigger intracellular second messenger cascades (cAMP, IP3/DAG, Ca). Effects are fast (seconds to minutes) and transient.

Feedback: almost every endocrine axis uses long-loop negative feedback. The end-product hormone suppresses the hypothalamus and pituitary that produced the upstream releasing and stimulating hormones. Cortisol suppresses CRH and ACTH. Thyroid hormone suppresses TRH and TSH. Disease often manifests when feedback is broken: primary failure (gland itself broken; downstream hormone low, upstream high because no negative feedback), secondary failure (pituitary broken; both upstream and downstream low), tertiary failure (hypothalamus broken). Pattern recognition here is high-yield.

Teaching This Topic

Before the video (drawing prompt)

Have students sketch the hypothalamus-pituitary-target gland axis as a feedback loop. Where does the end-hormone send its negative feedback signal?

Common misconceptions to address

- Students think hormones act on every cell. They act only on cells that have the right receptor.
- Steroid versus peptide mechanism is sometimes blurred. Steroids = intracellular receptor + transcription. Peptides = surface receptor + second messenger.
- Negative feedback in endocrine systems is sometimes thought to be optional. It is the default; positive feedback is the rare exception (LH surge before ovulation).

Order of operations (what to teach first)

Three chemical classes, then two mechanisms, then feedback patterns and the primary/secondary/tertiary failure logic.

Self-test prompts

- Where are steroid hormone receptors located?
- Why must insulin be injected and not swallowed?
- What does it mean when a patient has high ACTH and low cortisol?

Why This Matters to Your Patients

Primary hypothyroidism (Hashimoto): low T4, high TSH. Secondary hypothyroidism (pituitary failure): low T4, low TSH. Same low T4, different cause. Cushing syndrome: high cortisol producing hyperglycemia, central obesity, immunosuppression, hypertension. Addison disease: primary adrenal failure with low cortisol, high ACTH, skin hyperpigmentation. Diabetes insipidus: low ADH, dilute high-volume urine. SIADH: high ADH, concentrated low-volume urine, dilutional hyponatremia. Every one of these stories is a hormone level plus a feedback consequence.

TOPIC 2 OF 2 . WEEK 5 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Major Endocrine Glands

Pituitary, thyroid, parathyroid, adrenal, pancreas, gonads: who makes what and what it does.

The Science

The pituitary is the master, but the hypothalamus is its boss. The hypothalamus releases small peptides (CRH, TRH, GnRH, GHRH, etc.) that travel via a tiny portal system to the anterior pituitary, which then releases the trophic hormones: ACTH, TSH, FSH, LH, GH, prolactin. The posterior pituitary stores and releases ADH and oxytocin made in the hypothalamus.

Thyroid: T3 and T4 set metabolic rate; calcitonin lowers blood calcium (modest in adults). Parathyroids (four small glands posterior to the thyroid) release PTH, which raises blood calcium by osteoclast activation, renal calcium retention, and vitamin D activation. Adrenal cortex (three zones, top to bottom): zona glomerulosa makes aldosterone (Na retention), zona fasciculata makes cortisol (stress and metabolism), zona reticularis makes androgens. Adrenal medulla makes epinephrine and norepinephrine (sympathetic effector).

Pancreas islets: alpha cells make glucagon (raises blood glucose), beta cells make insulin (lowers blood glucose), delta cells make somatostatin (regulator). Gonads: testes make testosterone; ovaries make estrogen and progesterone. Each gland is a story; pair each hormone with its main job in one sentence and students can build a matrix they will use for the rest of healthcare.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a body outline and place every major endocrine gland with one hormone and one job per gland. They have to commit before the video.

Common misconceptions to address

- Students confuse hypothalamus and pituitary. Hypothalamus is the upstream controller; pituitary is the executive that fires hormones into the blood.
- ADH and oxytocin are sometimes confused. ADH = water retention; oxytocin = labor, milk letdown, social bonding.
- The adrenal cortex and medulla are different tissues with different embryonic origins, despite sharing one capsule.

Order of operations (what to teach first)

Hypothalamus-pituitary axis, then thyroid/parathyroid, then adrenals, then pancreas, then gonads.

Self-test prompts

- Name the gland that makes ADH and where it is released.
- What three things does PTH do to raise blood calcium?
- Which adrenal zone makes aldosterone and what does aldosterone do?

Why This Matters to Your Patients

Type 1 diabetes: autoimmune destruction of pancreatic beta cells; no insulin; hyperglycemia, ketoacidosis without exogenous insulin.

Type 2 diabetes: insulin resistance plus eventual beta cell failure. Hyperthyroidism (Graves): heat intolerance, weight loss, tremor, palpitations. Hypothyroidism: cold intolerance, weight gain, fatigue, bradycardia. Pheochromocytoma: a catecholamine-secreting adrenal tumor producing episodic hypertension, headache, sweating, palpitations. Hyperparathyroidism: high calcium producing kidney stones, bone loss, neuropsychiatric symptoms. Each gland has its overproduction and underproduction story; nurses need both for any patient with endocrine symptoms.

Module 10: Blood

Topics: Blood Composition and Hemopoiesis, Hemostasis and Blood Typing

TOPIC 1 OF 2 . WEEK 6 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Blood Composition and Hemopoiesis

Plasma, formed elements, and how blood cells are made in red marrow.

The Science

Blood is connective tissue: cells in a liquid matrix (plasma). About 55% plasma, 45% formed elements by volume. Plasma is mostly water plus proteins (albumin maintains oncotic pressure; globulins include antibodies; fibrinogen makes fibrin for clots), electrolytes, glucose, hormones, and wastes.

Formed elements: erythrocytes (RBCs, 99% of cells; biconcave, no nucleus, packed with hemoglobin, 120-day lifespan, function = O₂/CO₂ transport). Leukocytes (WBCs, immune cells). Platelets (cell fragments from megakaryocytes; central to hemostasis).

WBC types: granulocytes (neutrophils, eosinophils, basophils, granular cytoplasm) and agranulocytes (lymphocytes, monocytes). Neutrophils are the most abundant; first responders to bacterial infection. Eosinophils target parasites and allergic responses. Basophils release histamine in inflammation. Lymphocytes (B and T) drive adaptive immunity. Monocytes leave blood and mature into tissue macrophages.

Hematopoiesis: all blood cells come from hematopoietic stem cells in red marrow (axial skeleton and proximal limb bones in adults). Erythropoiesis (RBC production) is driven by erythropoietin, made by the kidneys in response to low oxygen. Low EPO (chronic kidney disease) = anemia.

Teaching This Topic

Before the video (drawing prompt)

Ask students to predict the percentages of plasma vs cells, and to list the five kinds of WBC in order of abundance. They have to commit first.

Common misconceptions to address

- Students think blood is just RBCs. By cell count yes, but by function the WBCs and platelets are equally critical.
- Blood is sometimes described as 'red liquid.' It is a connective tissue with a specific cell-to-matrix structure.
- RBCs without a nucleus seems like a disability. It is a specialization: more room for hemoglobin, more flexible to squeeze through capillaries.

Order of operations (what to teach first)

Plasma vs formed elements, then RBC specifics, then WBC types and roles, then platelets briefly, then hematopoiesis.

Self-test prompts

- What percentage of blood is plasma?
- Name the WBC types from most to least abundant.
- Why are patients with chronic kidney disease anemic?

Why This Matters to Your Patients

CBC (complete blood count) is the most common lab test. Low hemoglobin = anemia (many causes: iron deficiency, B12, chronic disease, blood loss). High WBC = infection or leukemia. Low platelets = bleeding risk. The differential (percentages of each WBC type) tells you what kind of infection: bacterial typically raises neutrophils; viral often raises lymphocytes; parasites raise eosinophils. Patients on chemotherapy have predictable cytopenias from bone marrow suppression; nurses time their assessments and protective measures accordingly. Recombinant EPO is given to dialysis patients to treat anemia of CKD.

TOPIC 2 OF 2 . WEEK 6 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Hemostasis and Blood Typing

Stopping bleeding in three steps, and the basics of ABO and Rh.

The Science

Three phases of hemostasis: vascular spasm (damaged vessel constricts), platelet plug formation (platelets adhere to exposed collagen via von Willebrand factor, activate, and recruit more platelets), and coagulation (the clotting cascade ends in fibrin, which reinforces the plug). The cascade has intrinsic and extrinsic pathways that converge on a common pathway producing thrombin, which converts fibrinogen to fibrin.

ABO typing: RBCs may carry A antigen, B antigen, both, or neither (Type O). Plasma carries antibodies against absent antigens (Type A has anti-B; Type O has both; Type AB has neither). Type O is the universal RBC donor (no antigens to trigger recipient antibodies); Type AB is the universal RBC recipient (no antibodies to attack donor RBCs). For plasma, the rule reverses: AB is the universal plasma donor.

Rh factor: Rh⁺ means D antigen present. Anti-Rh antibodies form only after exposure. Critical in pregnancy: an Rh⁻ mother carrying an Rh⁺ fetus may produce anti-Rh antibodies, which can cross the placenta in a future pregnancy and attack an Rh⁺ fetus, producing hemolytic disease of the newborn. RhoGAM (anti-Rh antibody given to the mother) prevents this by clearing fetal Rh⁺ cells before her immune system responds.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the three phases of hemostasis as a flowchart and predict what a missing platelet plug, or missing fibrin, would mean.

Common misconceptions to address

- Students think clotting and clot dissolution are separate things. They are balanced: fibrinolysis is happening alongside coagulation; the net result is the lifespan of the clot.
- ABO and Rh are sometimes mixed. ABO antibodies are pre-formed (natural); Rh antibodies form only after exposure.
- Type O is universal donor for RBCs, NOT for plasma.

Order of operations (what to teach first)

Three phases of hemostasis, then ABO logic, then Rh logic with the pregnancy implication.

Self-test prompts

- Name the three phases of hemostasis.
- Which is the universal RBC donor and which is the universal plasma donor?
- Why are Rh⁻ women given RhoGAM during pregnancy?

Why This Matters to Your Patients

Hemophilia A and B (clotting factor deficiencies) produce delayed bleeding into joints and tissue. Von Willebrand disease (the most common inherited bleeding disorder) produces mucocutaneous bleeding. Warfarin inhibits vitamin-K-dependent clotting factors and is used to prevent stroke in atrial fibrillation; the trade-off is bleeding risk, which is monitored with INR. Heparin inhibits thrombin and is used for acute anticoagulation. Transfusion reactions occur when ABO mismatch happens (recipient antibodies attack donor RBCs); hemolysis can be rapidly fatal, hence the two-nurse identity check at the bedside. Liver disease produces coagulopathy because the liver makes most clotting factors.

Module 11: Cardiovascular System

Topics: Heart Anatomy and the Cardiac Cycle, Cardiac Conduction System, Blood Vessels and Hemodynamics

TOPIC 1 OF 3 . WEEK 6 . 12 CARDS (5 DOK 1 / 3 DOK 2 / 4 DOK 3)

Heart Anatomy and the Cardiac Cycle

Chambers, valves, great vessels, and the pressure-volume story of one heartbeat.

The Science

The heart has four chambers and four valves. Right side carries deoxygenated blood: vena cavae into right atrium (RA) through tricuspid valve into right ventricle (RV), through pulmonary valve into pulmonary trunk and lungs. Left side carries oxygenated blood: pulmonary veins into left atrium (LA), through mitral (bicuspid) valve into left ventricle (LV), through aortic valve into aorta. The two sides are functionally in series.

The cardiac cycle has four phases (one beat = one cycle): ventricular filling (AV valves open, blood drops in passively, then atrial systole tops it off), isovolumetric contraction (all valves closed, pressure rises in the ventricles), ventricular ejection (semilunar valves open, blood is ejected), isovolumetric relaxation (all valves closed, pressure falls). The cycle then restarts.

Heart sounds: S1 ('lub') is the closure of the AV valves at the start of systole. S2 ('dub') is the closure of the semilunar valves at the end of systole. S3 (early diastolic gallop) can be normal in young athletes or pathologic in heart failure. S4 (late diastolic) is often pathologic, reflecting a stiff ventricle. Murmurs are audible turbulence; the timing in the cycle and where they are loudest tell you which valve is malfunctioning.

Frank-Starling mechanism: within the working range, the more the ventricle is stretched by incoming blood (preload), the more forcefully it contracts on the next beat. The heart pumps out what it receives. This is the basis of moment-to-moment matching of cardiac output to venous return.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the heart in cross-section, label all four chambers and four valves, and trace the path of a red blood cell from vena cava to aorta.

Common misconceptions to address

- Students think the heart is one pump. It is two pumps in series.
- S1 and S2 are sometimes mixed. Cue: S1 (lub) is louder and longer (AV valves shutting against ventricular pressure); S2 (dub) is sharper.
- Frank-Starling is sometimes thought of as a feedback loop. It is a structural property of muscle, not a regulatory loop.

Order of operations (what to teach first)

Anatomy and blood flow, then valves and the cardiac cycle phases, then heart sounds, then Frank-Starling as the integrating concept.

Self-test prompts

- Trace blood from the inferior vena cava to the aorta, naming every chamber and valve.
- Which valves close at S1? Which at S2?
- What is happening during isovolumetric contraction?

Why This Matters to Your Patients

Mitral regurgitation: blood leaks back into the LA during systole; forward cardiac output drops; chronic volume overload dilates the LV. Aortic stenosis: narrowed valve raises the pressure the LV must generate; chronic pressure overload causes concentric LV hypertrophy. Heart failure with reduced ejection fraction (HFrEF): the LV ejects less than 40% per beat; backward congestion produces pulmonary edema; forward underperfusion produces fatigue and renal dysfunction. Cardiac tamponade: fluid in the pericardial sac compresses the heart, dropping preload and cardiac output; this is a true emergency. Nurses monitor for the early signs (rising JVP, dropping BP, muffled heart sounds: Beck's triad). Every patient with a murmur deserves a careful reading of timing and location.

TOPIC 2 OF 3 . WEEK 6 . 8 CARDS (4 DOK 1 / 2 DOK 2 / 2 DOK 3)

Cardiac Conduction System

How the heart paces itself: pacemaker, delay, and rapid ventricular spread.

The Science

The heart paces itself. The sinoatrial (SA) node sits in the right atrial wall and fires 60-100 times per minute at rest. From there, the impulse spreads through atrial myocardium, hits the atrioventricular (AV) node, and is deliberately held there for about 100 milliseconds. That delay is the load-bearing concept: it lets the atria finish contracting and emptying before the ventricles fire. Without it, atria and ventricles would contract simultaneously, ventricular filling would collapse, and stroke volume would crash.

After the AV delay, the signal races through the Bundle of His, down the right and left bundle branches, and out into the Purkinje fibers, which distribute it almost simultaneously to the entire ventricular myocardium. That speed is essential: ventricles need to contract as a coordinated unit to eject effectively. Slow or uneven spread produces inefficient pumping.

For essentials scope, you do not have to teach ECG wave-by-wave interpretation. The core concept is the pathway and the timing. Mention arrhythmias as a category (AV block, atrial fibrillation, ventricular fibrillation) so students recognize the vocabulary, but you do not need them reading rhythm strips.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the heart and trace the conduction pathway with arrows from SA to ventricular myocardium. They commit before the video confirms the route.

Common misconceptions to address

- Students think the AV node speeds the signal up. It deliberately slows it; the delay is the entire point.
- The conduction system is sometimes confused with the coronary arteries. Conduction is electrical (myocardial cells specialized for fast conduction); arteries are the blood supply.
- Ventricular fibrillation is sometimes thought to be a fast heart rate. It is chaotic activity with no coordinated contraction at all; the ventricles quiver.

Order of operations (what to teach first)

Pacemaker (SA node) → AV delay (why it matters) → fast ventricular spread (Bundle of His, Purkinje) → coordinated ejection. Add arrhythmias as failure modes of each step.

Self-test prompts

- Without notes, draw the conduction pathway in order.
- Why is the AV nodal delay necessary?
- Why is rapid Purkinje conduction important for effective ejection?

Why This Matters to Your Patients

Atrial fibrillation is the most common sustained arrhythmia and a major stroke risk: stagnant blood in the fibrillating atrium can clot and embolize to the brain. Anticoagulation (warfarin or a DOAC) is standard. AV blocks (especially third-degree) drop cardiac output because atrial and ventricular contractions decouple; pacemakers restore coordination. Ventricular fibrillation is the cardiac-arrest rhythm that responds to defibrillation, which is why AEDs are everywhere. Every nurse on a telemetry floor is watching this pathway in real time.

TOPIC 3 OF 3 . WEEK 6 . 5 CARDS (3 DOK 1 / 1 DOK 2 / 1 DOK 3)

Blood Vessels and Hemodynamics

Vessel types, blood pressure determinants, and how flow is regulated.

The Science

Blood vessels are a structural and functional gradient. Arteries are thick-walled and elastic, carrying blood under high pressure. Arterioles are smaller and heavily muscled; they are the primary site of vascular resistance. Capillaries are single-endothelial-layer tubes where exchange happens. Venules and veins are thin-walled with valves; they hold most of the blood volume (capacitance vessels).

Blood pressure equation: $BP = \text{cardiac output (CO)} \times \text{total peripheral resistance (TPR)}$. Cardiac output equation: $CO = \text{stroke volume (SV)} \times \text{heart rate (HR)}$. Stroke volume is set by preload, contractility, and afterload. Total peripheral resistance is dominated by arteriolar diameter. Mean arterial pressure (MAP) is the perfusion pressure that matters to tissues; you want it roughly 65 or above.

Regulation: short-term, the baroreflex (carotid sinus and aortic arch baroreceptors to medullary cardiovascular centers) adjusts HR and vascular tone within seconds. Long-term, the kidneys via RAAS (renin-angiotensin-aldosterone) and ANP regulate blood volume. Both axes are clinical workhorses.

Teaching This Topic

Before the video (drawing prompt)

Have students predict what happens to BP if (a) arterioles vasoconstrict, (b) HR doubles, (c) plasma volume drops. Then they have to commit before checking.

Common misconceptions to address

- Students think 'high BP' just means HR is fast. It can also mean high resistance, high volume, high contractility.
- Veins are sometimes called 'low-pressure arteries.' They have different structure: thin walls, valves, more compliant.
- Capillaries are thought of as small arteries. They are unique: single layer, no muscle, designed for exchange.

Order of operations (what to teach first)

Vessel types, then the BP equation and components, then short-term and long-term regulation.

Self-test prompts

- Where in the circulation is the primary site of resistance?
- Write the equations for cardiac output and blood pressure.
- Why do veins have valves?

Why This Matters to Your Patients

Hypertension is the most common chronic disease in adults. Treatments target each node of the BP equation: diuretics drop volume (and therefore CO), beta blockers drop HR and contractility, calcium channel blockers drop contractility and vasodilate, ACE inhibitors drop RAAS-driven vasoconstriction and volume retention. Hypovolemic shock from blood loss produces tachycardia (baroreflex compensation) and vasoconstriction; nurses watch for these as the warning signs before BP itself collapses. Orthostatic hypotension (BP drops when standing) reflects autonomic or volume problems and is a fall risk in the elderly.

Module 12: Lymphatic and Immune System

Topics: Lymphatic System and Innate Immunity, Adaptive Immunity

TOPIC 1 OF 2 . WEEK 7 . 5 CARDS (3 DOK 1 / 1 DOK 2 / 1 DOK 3)

Lymphatic System and Innate Immunity

Lymph vessels, lymphoid organs, and the body's general (non-specific) defenses.

The Science

The lymphatic system returns interstitial fluid to the blood (about 3 L per day) and filters that fluid through lymph nodes for immune surveillance. Lymph vessels parallel the venous system. Major lymphoid organs: lymph nodes (filter lymph), spleen (filters blood), thymus (T cell maturation), tonsils, and MALT (mucosa-associated lymphoid tissue scattered along the gut, airway, etc.).

Innate immunity is the body's general (non-specific) defense and is the first line. Surface barriers: skin (stratified squamous keratinized), mucous membranes (with antimicrobial peptides and IgA), low stomach pH, ciliated airway. If something gets past, innate cells respond: neutrophils (first responders, eat bacteria), macrophages (eat and recruit), natural killer cells (kill virus-infected and tumor cells without prior exposure). The complement system is a cascade of plasma proteins that lyse pathogens and mark them for phagocytosis. Interferons are antiviral cytokines.

Inflammation: the response to tissue damage or infection. Vasodilation and increased vascular permeability deliver immune cells and plasma proteins to the site. Cardinal signs: redness (rubor), heat (calor), swelling (tumor), pain (dolor), loss of function. Chemotaxis recruits more cells. Inflammation is necessary but can cause tissue damage if uncontrolled.

Teaching This Topic

Before the video (drawing prompt)

Have students sketch the path of interstitial fluid into a lymphatic capillary, through a lymph node, and back to the blood. Then list the five cardinal signs of inflammation.

Common misconceptions to address

- Students confuse the lymphatic and the cardiovascular systems. They are parallel but separate; lymph empties into the venous system at the subclavian veins.
- Innate immunity is sometimes dismissed as primitive. It handles 99% of insults without you noticing.
- Inflammation is seen as pathological. It is the body's defense; chronic or uncontrolled inflammation is the pathology.

Order of operations (what to teach first)

Lymphatic anatomy and function, then innate immunity layers (barriers, cells, complement), then inflammation as the visible result.

Self-test prompts

- Name the four cardinal signs of inflammation.
- Name two phagocytic cells.
- Why do asplenic patients need extra vaccines and antibiotic precautions?

Why This Matters to Your Patients

Lymphedema (often after lymph node dissection for breast cancer surgery, or from filariasis) is the failure of lymph drainage; the affected limb swells permanently. Splenectomy patients are at high risk of overwhelming infection from encapsulated organisms (*Strep pneumoniae*, *H. influenzae*, *N. meningitidis*); they receive specific vaccines. Sepsis is uncontrolled systemic inflammation in response to infection; early recognition (SIRS criteria, qSOFA, lactate) is critical for survival. Allergies and anaphylaxis involve inappropriate or excessive innate (and adaptive) responses. Nurses caring for any patient with infection are watching for the transition from local to systemic inflammatory response.

TOPIC 2 OF 2 . WEEK 7 . 12 CARDS (5 DOK 1 / 3 DOK 2 / 4 DOK 3)

Adaptive Immunity

Specificity and memory: how B and T cells learn what to attack and remember it.

The Science

Adaptive immunity has two defining features: specificity (each lymphocyte recognizes one antigen) and memory (re-exposure produces a faster, larger response). It comes in two arms.

Humoral immunity: B cells, when activated, become plasma cells that produce antibodies. Antibodies (IgG, IgM, IgA, IgE, IgD) circulate and tag extracellular pathogens for destruction by neutralization, opsonization, complement activation, or antibody-dependent cell killing (ADCC). This is your primary defense against circulating bacteria, toxins, and viruses outside cells.

Cell-mediated immunity: T cells, in two main flavors. Helper T cells (CD4) coordinate the response by activating B cells, cytotoxic T cells, and macrophages. Cytotoxic T cells (CD8) directly kill infected or abnormal cells. T cells need antigen presented on MHC molecules: MHC class I (on all nucleated cells, presents intracellular peptides to CD8) or MHC class II (on antigen-presenting cells, presents extracellular peptides to CD4).

Activation requires two signals: antigen plus co-stimulation. This two-key rule prevents autoimmunity (a lymphocyte that binds self-antigen without co-stimulation is suppressed or killed). Clonal expansion produces effector cells (do the job) and memory cells (live for years, respond fast on re-exposure). Vaccination relies entirely on memory.

Teaching This Topic

Before the video (drawing prompt)

Have students predict: which arm of adaptive immunity is best at killing virus-infected cells versus extracellular bacteria? Then they have to commit.

Common misconceptions to address

- Students think innate and adaptive immunity work in series. They cooperate constantly; innate cells present antigen and drive adaptive responses.
- Antibodies are sometimes thought to kill pathogens directly. They tag, neutralize, or recruit; killing is usually by complement or phagocytes.
- Vaccines are sometimes thought to give you the disease. They give you the antigen without the pathogen; your memory cells will respond if you ever encounter the real one.

Order of operations (what to teach first)

Definition and features (specificity, memory), then humoral arm with antibody functions, then cell-mediated arm with MHC presentation, then activation rules and memory.

Self-test prompts

- Which T cell subset directly kills infected cells?
- Which MHC class presents to CD4 T cells?
- Why are vaccinated people protected even though they have never had the disease?

Why This Matters to Your Patients

HIV destroys CD4 T cells; without coordination, both humoral and cell-mediated immunity collapse, and patients develop opportunistic infections. Transplant patients receive immunosuppression to prevent T cell rejection of the graft; the trade-off is infection and cancer risk. Autoimmune diseases (lupus, rheumatoid arthritis, type 1 diabetes, MS, Hashimoto, Graves) reflect loss of self-tolerance. Vaccination has transformed pediatric mortality. Monoclonal antibody therapies (mAbs ending in -mab) target specific receptors or pathogens. Every patient with infection, autoimmunity, or immunodeficiency is a story about the adaptive immune system functioning, failing, or overshooting.

Module 13: Respiratory System

Topics: Respiratory Anatomy and Mechanics, Gas Exchange and Transport

TOPIC 1 OF 2 . WEEK 7 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Respiratory Anatomy and Mechanics

Conducting vs respiratory zones, and how pressure changes move air in and out.

The Science

Two zones. Conducting zone: nose, pharynx, larynx, trachea, bronchi, bronchioles. Warms, humidifies, filters. No gas exchange happens here. Respiratory zone: respiratory bronchioles, alveolar ducts, alveoli. Gas exchange happens here. The alveoli are simple squamous (Type I pneumocytes) with surfactant from Type II pneumocytes.

Pleura: parietal pleura lines the thoracic cavity; visceral pleura covers the lung. Between them is a tiny pleural cavity with serous fluid and (critically) negative pressure. The negative pressure couples the lung to the chest wall: as the chest wall expands during inspiration, the lung is pulled outward with it. Lose the negative pressure (pneumothorax) and the lung collapses elastically.

Mechanics: Boyle's law. Pressure and volume vary inversely at constant temperature. Inspiration: diaphragm contracts (pulls down) and external intercostals lift the rib cage; thoracic volume rises; alveolar pressure drops below atmospheric; air flows in. Expiration at rest is passive: muscles relax, elastic recoil shrinks the thorax, alveolar pressure rises above atmospheric, air flows out. Forced expiration adds internal intercostals and abdominal muscles.

Teaching This Topic

Before the video (drawing prompt)

Have students sketch the airway from mouth to alveolus, label conducting vs respiratory zone, and identify Type I and Type II pneumocytes at the alveolus.

Common misconceptions to address

- Students think the lungs actively pull air in. Air is sucked in passively by the pressure gradient produced by the diaphragm and chest wall.
- Surfactant is thought of as optional. It is essential: without it, alveolar surface tension collapses the alveoli.
- Pleural pressure is sometimes assumed to equal atmospheric. It is normally slightly negative; that is what holds the lung expanded.

Order of operations (what to teach first)

Anatomy and the two zones, then the pleura and the negative pressure rule, then Boyle's law and the mechanics of breathing.

Self-test prompts

- What is the main muscle of inspiration?
- What does surfactant do?
- Why does a pneumothorax cause the lung to collapse?

Why This Matters to Your Patients

Pneumothorax is air in the pleural cavity (from trauma, spontaneous rupture, or iatrogenic); the affected lung collapses; treatment is chest tube drainage to restore negative pressure. Pleural effusion is fluid in the pleural cavity (transudate from heart failure or hypoalbuminemia, exudate from infection or malignancy). Asthma is bronchospasm and airway inflammation, narrowing the conducting zone. COPD (emphysema and chronic bronchitis) destroys alveoli and obstructs airways. Neonatal respiratory distress syndrome is surfactant deficiency in premature infants; exogenous surfactant therapy is life-saving. Every breath in your patient is a Boyle's-law transaction.

TOPIC 2 OF 2 . WEEK 7 . 12 CARDS (4 DOK 1 / 4 DOK 2 / 4 DOK 3)

Gas Exchange and Transport

Diffusion at the alveoli and tissues, and how O₂ and CO₂ ride in the blood.

The Science

Gas exchange happens by diffusion down partial pressure gradients. External respiration: between alveolar air and pulmonary capillary blood. Alveolar O₂ is about 100 mmHg; pulmonary venous blood arrives at 40 mmHg; O₂ diffuses into blood until equilibrium. CO₂ goes the other way. Internal respiration: between systemic capillary blood and tissue. The gradients are reversed there.

Oxygen transport: about 98% bound to hemoglobin, 2% dissolved in plasma. Each Hb molecule binds 4 O₂ cooperatively (the binding of one O₂ increases affinity for the next). The Hb-O₂ dissociation curve is sigmoidal. Conditions that shift it right (acidosis, high CO₂, high temperature, high 2,3-BPG: the Bohr effect) cause Hb to release MORE O₂ to tissues. This is why exercising muscle, which is acidic, hot, and high in CO₂, gets extra oxygen automatically.

CO₂ transport: about 70% as bicarbonate (the carbonic anhydrase reaction in RBCs: $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3 \rightarrow \text{H}^+ + \text{HCO}_3^-$), 20% bound to hemoglobin, 10% dissolved. The bicarbonate system is also the body's main blood buffer; respiratory rate adjusts CO₂ and therefore pH on a minute-to-minute basis.

Control of ventilation: central chemoreceptors in the medulla respond to CSF pH (a proxy for arterial CO₂). Peripheral chemoreceptors (carotid and aortic bodies) respond to severely low arterial O₂ (typically below 60 mmHg). CO₂ is the primary moment-to-moment driver. In chronic CO₂ retainers (severe COPD), the central drive desensitizes and the patient relies on the hypoxic drive; high inspired O₂ can blunt this and cause hypoventilation.

Teaching This Topic

Before the video (drawing prompt)

Have students predict: at exercising muscle, does Hb release more or less O₂? Why? They have to commit before the Bohr effect is explained.

Common misconceptions to address

- Students think dissolved O₂ is the main carrier. No, 98% is on Hb. Dissolved O₂ (measured as pO₂) is just the part that determines Hb saturation.
- CO₂ is thought to be transported mainly bound to Hb. No, mostly as bicarbonate.
- Students think low O₂ is the main driver of breathing. It is not. CO₂ (via pH of the CSF) is the primary minute-to-minute driver.

Order of operations (what to teach first)

Where exchange happens (external vs internal), then O₂ transport with the Bohr effect, then CO₂ transport with the bicarbonate buffer, then control of ventilation.

Self-test prompts

- What is the main form of CO₂ transport in blood?
- What conditions shift the Hb-O₂ curve to the right and what is the consequence?
- Why does CO poisoning cause hypoxia even though arterial pO₂ is normal?

Why This Matters to Your Patients

Carbon monoxide poisoning: CO binds Hb with 250x O₂ affinity; oxygen-carrying capacity collapses, even though pO₂ looks normal on blood gas. Pulse oximetry can be falsely normal; co-oximetry is required. Treatment is 100% O₂ and, in severe cases, hyperbaric oxygen. Acute respiratory acidosis (CO₂ retention from respiratory failure) produces falling pH; renal compensation takes days. Chronic respiratory acidosis (COPD) has compensated pH because the kidneys have retained HCO₃⁻ over time. Pulse oximetry, arterial blood gas interpretation, and supplemental O₂ titration are core nursing skills that depend on this physiology.

Module 14: Digestive System

Topics: GI Tract Anatomy and Motility, Digestion and Absorption

TOPIC 1 OF 2 . WEEK 7 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

GI Tract Anatomy and Motility

From mouth to anus, the layers of the gut wall, and how food moves.

The Science

The GI tract is a single tube from mouth to anus, about 9 meters long. Path: mouth, pharynx, esophagus, stomach, small intestine (duodenum, jejunum, ileum), large intestine (cecum, ascending, transverse, descending, sigmoid colon), rectum, anus. Accessory organs (salivary glands, liver, gallbladder, pancreas) deliver enzymes and bile into the tube.

Wall layers (deep to superficial): mucosa (epithelium, lamina propria, muscularis mucosae), submucosa (vessels, submucosal/Meissner plexus), muscularis externa (inner circular and outer longitudinal smooth muscle layers with myenteric/Auerbach plexus between), serosa or adventitia. The two enteric nerve plexuses make up the enteric nervous system, sometimes called the 'second brain' for its independent function.

Motility comes in two flavors. Peristalsis: a wave of smooth muscle contraction that propels content forward. Segmentation: localized contractions that mix without net forward motion (small intestine). Sphincters divide the tract into functional compartments: lower esophageal (cardia), pyloric (stomach to duodenum), ileocecal (ileum to cecum), internal and external anal.

Teaching This Topic

Before the video (drawing prompt)

Have students trace the GI tract from mouth to anus, naming every segment in order, then label the major sphincters.

Common misconceptions to address

- Students think the small intestine is small. It is small in diameter but very long (about 6 meters); the large intestine is shorter (about 1.5 m) but wider.
- Peristalsis and segmentation are sometimes mixed. Peristalsis moves food forward; segmentation mixes it in place.
- Sphincters are thought of as decorative. Each is functionally critical; failure produces specific clinical syndromes (achalasia, gastroparesis, anal incontinence).

Order of operations (what to teach first)

Organ sequence, then the wall layer structure (same throughout, with regional modifications), then motility and sphincters.

Self-test prompts

- Name the three parts of the small intestine in order.
- What is the function of the pyloric sphincter?
- How does peristalsis differ from segmentation?

Why This Matters to Your Patients

Achalasia: the lower esophageal sphincter cannot relax; food and liquid accumulate in the lower esophagus, producing dysphagia and regurgitation. Gastroparesis: delayed stomach emptying, often in diabetics from autonomic neuropathy. Small bowel obstruction: peristalsis pushes against a blockage, causing cramping, distension, vomiting. C. difficile colitis: pseudomembranous colon inflammation after antibiotics; severe diarrhea, electrolyte disturbance. Hirschsprung disease: absence of enteric plexus in part of the colon; obstruction in newborns. Every GI symptom should make you ask which segment is failing and how.

TOPIC 2 OF 2 . WEEK 7 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Digestion and Absorption

Where each macromolecule gets broken down and how absorption happens.

The Science

Each macromolecule has a digestion story. Carbohydrates: salivary amylase begins starch breakdown in the mouth (stops in the stomach because of acid). Pancreatic amylase resumes in the small intestine, producing disaccharides. Brush-border enzymes (maltase, sucrase, lactase) finish the job to monosaccharides, which are absorbed by SGLT1 (Na-glucose symporter) and GLUT2.

Proteins: pepsin in the stomach (activated from pepsinogen by acid) begins. Pancreatic proteases (trypsin, chymotrypsin, carboxypeptidase, activated by enterokinase in the duodenum) continue. Brush-border peptidases finish to amino acids and small

peptides, absorbed by amino-acid transporters.

Lipids: bile (made by liver, stored in gallbladder, released into duodenum in response to CCK) emulsifies fats. Pancreatic lipase hydrolyzes triglycerides to monoglycerides and fatty acids. Products form micelles, diffuse into enterocytes, reassemble into chylomicrons, and enter lymph via lacteals. Fat-soluble vitamins (A, D, E, K) ride along.

Absorptive surface: plicae circulares (folds), villi (finger-like projections), microvilli (brush border). Together they multiply surface area roughly 600-fold compared to a smooth tube of the same length. Small intestine = absorption central. Large intestine reclaims water and electrolytes, ferments residual carbohydrates, and forms feces.

Teaching This Topic

Before the video (drawing prompt)

Have students predict where each macromolecule (carbs, proteins, fats) is digested and where it is absorbed. They have to commit first.

Common misconceptions to address

- Students think the stomach does most of the digestion. The stomach starts protein digestion and stores food; the small intestine does the lion's share.
- Fats are thought to enter the blood like carbs and protein. They enter lymph first (chylomicrons via lacteals).
- Bile is sometimes confused with a digestive enzyme. It is not an enzyme; it is an emulsifier that lets enzymes (lipase) access the substrate.

Order of operations (what to teach first)

Each macromolecule's digestion and absorption in turn, then the absorptive surface design as the unifying answer to 'how does so much get absorbed so fast.'

Self-test prompts

- Where is most absorption accomplished?
- What does bile do, and where does it come from?
- Why do fats enter lymph rather than blood directly?

Why This Matters to Your Patients

Lactose intolerance: lactase deficiency; lactose is not digested, ferments in the colon, draws water in osmotically, causing gas and diarrhea. Celiac disease: gluten triggers autoimmune destruction of small intestinal villi; absorption fails; the patient develops nutritional deficiencies. Cystic fibrosis affects the pancreas: thick secretions block enzyme delivery, causing maldigestion. Cholecystectomy (gallbladder removal): bile flow continues but is no longer concentrated and stored; patients often tolerate small fatty meals but not large ones. Bariatric surgery alters absorption and digestion deliberately for weight loss; lifelong vitamin and protein monitoring is essential. GI complaints should prompt thinking about which step of digestion or absorption is failing.

Module 15: Urinary System and Fluid Balance

Topics: Kidney Anatomy and Glomerular Filtration, Tubular Function and Urine Concentration, Fluid, Electrolyte, and Acid-Base Balance

TOPIC 1 OF 3 . WEEK 8 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Kidney Anatomy and Glomerular Filtration

From renal cortex to nephron to filtration barrier: how urine starts.

The Science

The kidney's gross anatomy is layered: outer cortex, inner medulla (with renal pyramids), drain to the renal pelvis to the ureter. Each kidney has about a million nephrons. Blood path: renal artery to interlobar arteries to arcuate to interlobular to afferent arteriole into the glomerulus, then efferent arteriole out, then peritubular capillaries (or vasa recta for juxtamedullary nephrons), and venous return.

The nephron is the functional unit. It has the renal corpuscle (glomerulus surrounded by Bowman's capsule) and the tubule (proximal convoluted tubule, loop of Henle, distal convoluted tubule, collecting duct). Cortical nephrons (85%) have short loops of Henle. Juxtamedullary nephrons (15%) have long loops that dip deep into the medulla and produce the concentration gradient that allows the kidney to make concentrated urine.

Filtration: the filtration barrier has three layers (fenestrated capillary endothelium, basement membrane, podocyte slit diaphragms). Cells and large proteins are excluded; water, ions, glucose, amino acids, and small molecules pass freely. GFR (glomerular filtration rate) is about 125 mL/min, or 180 L per day. Of that, less than 1% becomes urine; the rest is reabsorbed. GFR is autoregulated by myogenic response (afferent arteriole constricts in response to stretch) and tubuloglomerular feedback (macula densa senses tubular flow).

Teaching This Topic

Before the video (drawing prompt)

Have students draw a nephron with all its tubular segments and the vascular supply, then label the filtration barrier.

Common misconceptions to address

- Students think the kidney just makes urine. It also regulates BP (RAAS), acid-base balance, calcium, EPO production, vitamin D activation.
- Glomerular filtrate is sometimes confused with urine. Filtrate is roughly the composition of plasma minus proteins; urine is what is left after reabsorption.
- Tubular handling is thought of as passive. Most is active and tightly regulated.

Order of operations (what to teach first)

Gross anatomy, then the nephron structure, then the filtration barrier and GFR, then autoregulation.

Self-test prompts

- What is the typical GFR?
- Name the three layers of the glomerular filtration barrier.
- How does the kidney maintain a stable GFR despite changes in BP?

Why This Matters to Your Patients

Proteinuria suggests damage to the filtration barrier (often at the podocyte slit diaphragm in nephrotic syndrome). Hematuria can mean glomerular (RBC casts on urinalysis) or post-glomerular (stones, infection) bleeding. Acute kidney injury (AKI) can be prerenal (poor perfusion), intrinsic (the kidney tissue is damaged), or postrenal (obstruction). Chronic kidney disease (CKD) stages are based on GFR; treatment escalates from medical management to dialysis to transplant. Every patient with deranged labs (creatinine, BUN, electrolytes, acid-base) deserves a kidney-centric reading.

TOPIC 2 OF 3 . WEEK 8 . 12 CARDS (5 DOK 1 / 3 DOK 2 / 4 DOK 3)

Tubular Function and Urine Concentration

How the tubules reshape the filtrate into a final urine, and how ADH and aldosterone fine-tune it.

The Science

The tubule reshapes the filtrate. Each segment has characteristic transport biology, and each is a drug target.

Proximal convoluted tubule (PCT): bulk reabsorption. About 65% of filtered Na, water, glucose, amino acids, and bicarbonate are reabsorbed here. Glucose has a tubular maximum (T_m); above blood glucose of about 180 mg/dL the transporters saturate and glucose

spills into urine (glucosuria of diabetes). Loop of Henle: descending limb is permeable to water but not ions (water leaves, filtrate concentrates); thick ascending limb is impermeable to water but actively reabsorbs Na, K, 2Cl via the NKCC2 cotransporter (target of loop diuretics like furosemide). Net effect: a hypertonic medullary interstitium (the countercurrent multiplier).

Distal convoluted tubule (DCT): reabsorbs Na and Cl via the NCC cotransporter (target of thiazide diuretics). Late DCT and collecting duct: principal cells under aldosterone control reabsorb Na and secrete K (target of potassium-sparing diuretics like spironolactone). Intercalated cells handle acid-base by secreting H⁺ or bicarbonate.

Hormonal control: ADH (from posterior pituitary, rises when plasma osmolarity is high or volume is low) inserts aquaporin-2 channels in the collecting duct apical membrane; water flows out of the duct into the hypertonic medulla, producing concentrated urine.

Aldosterone (from adrenal cortex) drives Na reabsorption (water follows) and K secretion in the collecting duct. ANP (from atria, in volume overload) opposes RAAS and promotes Na and water excretion.

Teaching This Topic

Before the video (drawing prompt)

Have students label a nephron with each segment's job: 'PCT does X, loop does Y, DCT does Z, collecting duct does W.' They have to commit before video.

Common misconceptions to address

- Students think the collecting duct is what concentrates the urine. The collecting duct uses the gradient that the loop of Henle established; both are required.
- Aldosterone is sometimes confused with ADH. Cue: aldosterone moves SALT (and water follows); ADH moves WATER (alone).
- Diuretics are not all the same; loop, thiazide, and K-sparing have different sites and different electrolyte profiles.

Order of operations (what to teach first)

Each tubular segment in order with its main job, then the countercurrent multiplier, then ADH and aldosterone as the fine-tuning controls.

Self-test prompts

- Which segment reabsorbs the most filtered Na?
- What does ADH do mechanistically in the collecting duct?
- How does the countercurrent multiplier set up urine concentration?

Why This Matters to Your Patients

Diabetes insipidus: ADH absent (central) or ineffective (nephrogenic). Result: large volumes of dilute urine, thirst, risk of hypernatremia. Treatment: desmopressin (synthetic ADH) or address the cause. SIADH: too much ADH. Result: small volumes of concentrated urine, water retention, dilutional hyponatremia. Furosemide blocks NKCC2 in the loop; potent diuresis with hypokalemia, hypocalcemia risk. Thiazides block NCC in the DCT; milder diuresis. Spironolactone blocks aldosterone receptor; K-sparing, used in heart failure and primary hyperaldosteronism. Nurses titrating fluids and electrolytes in CHF, cirrhosis, and renal failure are working this physiology constantly.

TOPIC 3 OF 3 . WEEK 8 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Fluid, Electrolyte, and Acid-Base Balance

Body fluid compartments, the main electrolytes, and how the body keeps pH near 7.4.

The Science

Body fluid compartments: total body water is about 60% of body weight. Intracellular fluid (ICF) is about two-thirds; extracellular fluid (ECF) is about one-third. ECF subdivides into interstitial (75%) and plasma (25%). The compartments have very different electrolyte profiles: ICF is high K and low Na, while ECF is the reverse. These differences are maintained by the Na/K ATPase and matter because shifts can be catastrophic (hyperkalemia causes lethal arrhythmias because it disrupts the membrane potential).

Electrolyte priorities for the boards and the bedside: Na sets ECF osmolarity and volume; hyponatremia and hypernatremia produce neurologic symptoms. K sets membrane excitability; hyperkalemia and hypokalemia produce cardiac and muscle symptoms. Calcium has neuromuscular effects and is regulated by PTH, calcitonin, and vitamin D. Magnesium and phosphate are quieter but important cofactors and bone components.

Acid-base: normal arterial pH is 7.35-7.45. Three lines of defense. Buffers (bicarbonate, phosphate, protein) blunt acid loads on the second-to-second scale. Respiratory compensation adjusts CO₂ within minutes. Renal compensation adjusts HCO₃⁻ and H⁺ excretion over hours to days. Four classic disturbances: respiratory acidosis (high CO₂), respiratory alkalosis (low CO₂), metabolic acidosis (low HCO₃⁻ or high non-CO₂ acid), metabolic alkalosis (high HCO₃⁻ or loss of acid). Pattern recognition is the entire game: pH, pCO₂,

HCO₃⁻ read together tell the story.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the body fluid compartments as nested boxes with percentages, then label which compartment is high in K versus high in Na.

Common misconceptions to address

- Students think 'edema' is too much total water. Often it is redistribution between compartments (low albumin lets fluid leak from plasma to interstitium).
- Hyponatremia is sometimes treated as low total body Na. Often it is too much water, not too little salt.
- Acid-base is reduced to memorizing equations. The story is which system is broken and which is compensating.

Order of operations (what to teach first)

Compartments and their composition, then key electrolytes and their disturbances, then acid-base with the four-disturbance pattern.

Self-test prompts

- Which compartment holds the most water?
- What does the pattern pH 7.20, pCO₂ 25, HCO₃⁻ 12 represent?
- Why does severe vomiting cause metabolic alkalosis?

Why This Matters to Your Patients

Diabetic ketoacidosis: ketoacids accumulate, HCO₃⁻ is consumed buffering them, pH drops, the patient hyperventilates (Kussmaul) to drop CO₂ in compensation. Vomiting: loss of gastric H⁺ causes metabolic alkalosis; volume loss adds aldosterone-driven K loss. COPD: chronic CO₂ retention with renal compensation (high HCO₃⁻) and chronic compensated respiratory acidosis. Hyperkalemia (renal failure, K-sparing diuretics, Addison) is a true emergency because it causes peaked T waves and ventricular arrhythmias. Hyponatremia in elderly patients on thiazides is common and can cause seizures. Every set of labs has a fluid-and-electrolyte story; nurses who can read it catch problems early.

Module 16: Reproductive System

Topics: Male Reproductive System, Female Reproductive System, Pregnancy A&P (Basics)

TOPIC 1 OF 3 . WEEK 8 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Male Reproductive System

Testis structure, spermatogenesis, accessory glands, and the hormonal axis.

The Science

Testes do two jobs: produce sperm and produce testosterone. Sperm production happens in the seminiferous tubules, which are lined with Sertoli cells that support developing spermatogonia through meiosis. Testosterone is produced by Leydig cells (interstitial cells) outside the tubules. The blood-testis barrier (formed by Sertoli cell tight junctions) protects developing sperm from immune attack.

Sperm pathway: seminiferous tubule, rete testis, epididymis (sperm matures here, becomes motile, is stored), vas deferens (ascending through the inguinal canal), ejaculatory duct (joining with seminal vesicle duct), prostatic urethra, urethra. Accessory glands add to the seminal fluid: seminal vesicles (fructose, prostaglandins, about 60% of volume), prostate (alkaline secretion, enzymes, about 30%), bulbourethral glands (mucus pre-ejaculate).

Hormonal axis: hypothalamic GnRH drives pituitary LH and FSH. LH stimulates Leydig cells to make testosterone. FSH plus testosterone supports Sertoli cells and spermatogenesis. Testosterone feeds back negatively on the hypothalamus and pituitary to stabilize the system. Inhibin from Sertoli cells specifically suppresses FSH, fine-tuning spermatogenesis.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a cross-section of a testis with seminiferous tubules and Leydig cells labeled, then trace the path of sperm from tubule to urethra.

Common misconceptions to address

- Students think the testis is one structure with one function. It does sperm production AND testosterone production via different cells.
- The epididymis is sometimes overlooked. Sperm leaving the seminiferous tubules are not yet motile; they mature in the epididymis.
- Testosterone is thought of as 'the male hormone' but it is also made (in smaller amounts) by the adrenal cortex in both sexes.

Order of operations (what to teach first)

Testis structure and its two jobs, then sperm pathway with accessory glands, then the hormonal axis.

Self-test prompts

- Which cells produce testosterone?
- Which cells support developing sperm?
- Why do anabolic steroid users develop testicular atrophy?

Why This Matters to Your Patients

Testicular torsion is a surgical emergency; twisting of the spermatic cord cuts off blood supply and the testis dies within hours. Varicocele (enlarged pampiniform plexus) is a common cause of infertility because the heat impairs spermatogenesis. Benign prostatic hyperplasia (BPH) is age-related prostate enlargement causing urinary symptoms. Prostate cancer is the most common solid tumor in men; PSA screening is controversial but widely used. Hypogonadism (from various causes) produces low testosterone, low energy, reduced libido, reduced muscle mass; replacement therapy is increasingly common but has risks.

TOPIC 2 OF 3 . WEEK 8 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Female Reproductive System

Ovary, oogenesis, the ovarian and uterine cycles, and the hormonal choreography.

The Science

Ovaries produce eggs (oocytes) and hormones. Females are born with all their primordial follicles already arrested in prophase I of meiosis. Each cycle a cohort of follicles begins maturing; usually one becomes dominant. The dominant follicle's granulosa and theca cells produce estrogen under FSH and LH stimulation.

Ovarian cycle (about 28 days). Follicular phase (days 1-14): follicle matures; estrogen rises. Ovulation (day 14): LH surge triggers release of the secondary oocyte. Luteal phase (days 15-28): ruptured follicle becomes the corpus luteum, which produces progesterone (and some estrogen) to support potential implantation. If no pregnancy, the corpus luteum regresses, hormones drop, and the cycle restarts.

Uterine (menstrual) cycle parallels the ovarian: menstrual phase (days 1-5, shedding of endometrium), proliferative phase (days 6-14, estrogen rebuilds endometrium), secretory phase (days 15-28, progesterone makes endometrium glandular and receptive). If no implantation, progesterone falls and the cycle restarts.

Hormonal control: GnRH from hypothalamus drives LH and FSH. Estrogen and progesterone feed back, mostly negatively. The key twist: rising estrogen mid-cycle switches feedback from negative to positive, triggering the LH surge. After ovulation, the corpus luteum's progesterone restores negative feedback. If pregnancy occurs, the embryo's hCG keeps the corpus luteum alive until the placenta takes over hormone production.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the ovarian cycle phases with FSH, LH, estrogen, and progesterone curves over 28 days. Then label the uterine phases underneath.

Common misconceptions to address

- Students think eggs are made monthly. All eggs are present from birth; the monthly cycle is which one matures.
- Estrogen is thought to only feed back negatively. The mid-cycle switch to positive feedback is the key trigger for ovulation.
- The menstrual cycle is reduced to the bleeding phase. It is the integration of ovarian and uterine cycles over the entire month.

Order of operations (what to teach first)

Ovarian anatomy and oogenesis, then the ovarian cycle, then the uterine cycle in parallel, then hormonal control with the feedback switch.

Self-test prompts

- What triggers ovulation?
- What does the corpus luteum produce?
- How do combined oral contraceptives prevent pregnancy?

Why This Matters to Your Patients

Polycystic ovary syndrome (PCOS): irregular cycles, anovulation, hyperandrogenism, insulin resistance; common cause of infertility. Endometriosis: endometrial tissue outside the uterus producing painful, cyclic bleeding and infertility. Menopause: ovarian estrogen production declines; hot flashes, bone loss, mood changes. Hormonal contraception, fertility treatment, and management of menstrual disorders all depend on understanding the axis. Pap smears, mammograms, and pelvic exams are screening tools nurses help administer. Pregnancy testing detects hCG.

TOPIC 3 OF 3 . WEEK 8 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Pregnancy A&P (Basics)

Fertilization, implantation, and the maternal-fetal interface. Maternal physiology highlights through the trimesters.

The Science

Fertilization happens in the ampulla of the uterine tube within hours of ovulation. The zygote divides as it travels toward the uterus and arrives as a blastocyst. Implantation in the endometrium is around day 6-7. Trophoblast cells secrete hCG almost immediately; hCG keeps the corpus luteum alive so it continues producing the progesterone that holds the endometrium in place. By the end of the first trimester, the placenta takes over hormone production.

The placenta is the working interface between mother and fetus. It exchanges gases, nutrients, and waste, but maternal and fetal blood never directly mix. It is also an endocrine organ (progesterone, estrogen, hPL) and an imperfect filter (many drugs, alcohol, and infections cross to the fetus).

Maternal physiology changes substantially. Blood volume rises about 40 percent by the second trimester. Cardiac output increases. Renal filtration goes up. The respiratory drive shifts because progesterone affects ventilation. Late in pregnancy, the enlarged uterus presses on the inferior vena cava when the mother lies supine, compressing venous return and dropping blood pressure: classic supine hypotensive syndrome of pregnancy. For essentials scope, skip the developmental biology details (germ layers, organogenesis week-by-week) and focus on the maternal-fetal interface.

Teaching This Topic

Before the video (drawing prompt)

Ask students to draw the maternal-fetal interface: maternal blood on one side, fetal vessels on the other, the placental membrane between. They have to commit to whether blood mixes (it does not) before the video confirms.

Common misconceptions to address

- Maternal and fetal blood mix at the placenta. They do not; only molecules diffuse across the placental membrane.
- Students think pregnancy is mostly about the baby. Maternal physiology changes dramatically: blood volume, cardiac output, renal function, respiratory drive.
- hCG is sometimes thought to come from the mother. It comes from trophoblast (fetal-side) tissue, which is why it is such a specific pregnancy marker.

Order of operations (what to teach first)

Fertilization and implantation → hormonal support (hCG, corpus luteum, placenta) → placental functions → maternal physiology by trimester.

Self-test prompts

- Where does fertilization typically occur?
- What is the role of hCG in early pregnancy?
- Why does maternal blood volume rise during pregnancy?

Why This Matters to Your Patients

Ectopic pregnancy: implantation outside the uterus, usually in the tube. Rupture produces hemorrhagic shock; this is a surgical emergency. Preeclampsia: third-trimester hypertension and proteinuria, with risk of progression to eclampsia (seizures); blood pressure monitoring and timely delivery are the management. Gestational diabetes: places mother at risk of preeclampsia and fetus at risk of macrosomia and shoulder dystocia at delivery. Supine hypotensive syndrome: a third-trimester patient becomes dizzy lying flat. Roll her left to relieve IVC compression. These are the prenatal nursing scenarios this unit prepares students for.